Database

Managing and querying structured and semi-structured data

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Amazon DynamoDB

NoSQL Serverless Database

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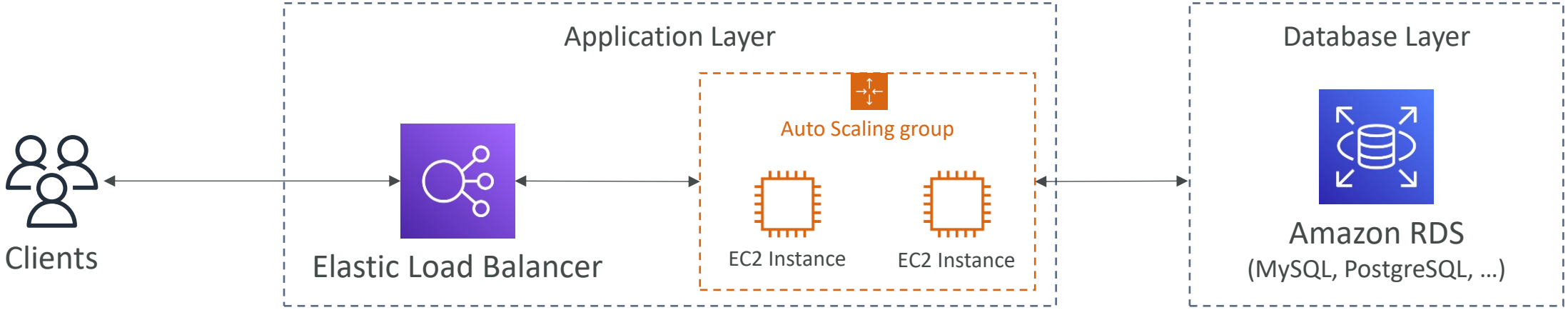
Traditional Architecture



- Traditional applications leverage RDBMS databases
- These databases have the SQL query language
- Strong requirements about how the data should be modeled
- Ability to do query joins, aggregations, complex computations
- Vertical scaling (getting a more powerful CPU / RAM / IO)
- Horizontal scaling (increasing reading capability by adding EC2 / RDS Read Replicas)

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NoSQL databases

- NoSQL databases are non-relational databases and are **distributed**
- NoSQL databases include MongoDB, DynamoDB, …
- NoSQL databases do not support query joins (or just limited support)
- All the data that is needed for a query is present in one row
- NoSQL databases don't perform aggregations such as "SUM", "AVG", …
- **NoSQL databases scale horizontally**

- There's no "right or wrong" for NoSQL vs SQL, they just require to model the data differently and think about user queries differently

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Amazon DynamoDB

- Fully managed, highly available with replication across multiple AZs
- NoSQL database - not a relational database
- Scales to massive workloads, distributed database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance (low latency on retrieval)
- Integrated with IAM for security, authorization and administration
- Enables event driven programming with DynamoDB Streams
- Low cost and auto-scaling capabilities
- Standard & Infrequent Access (IA) Table Class

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DynamoDB - Basics

- DynamoDB is made of **Tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of an item is **400KB**
- Data types supported are:
  - **Scalar Types** – String, Number, Binary, Boolean, Null
  - **Document Types** – List, Map
  - **Set Types** – String Set, Number Set, Binary Set

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DynamoDB – Primary Keys

- **Option 1: Partition Key (HASH)**
  - Partition key must be unique for each item
  - Partition key must be "diverse" so that the data is distributed
  - Example: "User_ID" for a users table

Primary Key | Attributes

Partition Key

User_ID	First_Name	Last_Name	Age
7791a3d6-…	John	William	46
873e0634-…	Oliver		24
a80f73a1-…	Katie	Lucas	31

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DynamoDB – Primary Keys

- **Option 2: Partition Key + Sort Key (HASH + RANGE)**
  - The combination must be unique for each item
  - Data is grouped by partition key
  - Example: users-games table, “User_ID” for Partition Key and “Game_ID” for Sort Key

Primary Key		Attributes	
Partition Key	Sort Key		
User_ID	Game_ID	Score	Result
7791a3d6-…	4421	92	Win
873e0634-…	1894	14	Lose
873e0634-…	4521	77	Win

Same partition key
Different sort key

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DynamoDB – Partition Keys (Exercise)

- We're building a movie database
- What is the best Partition Key to maximize data distribution?
  - movie_id
  - producer_name
  - leader_actor_name
  - movie_language

- "movie_id" has the highest cardinality so it's a good candidate
- "movie_language" doesn't take many values and may be skewed towards English so it's not a great choice for the Partition Key

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DynamoDB in Big Data

- Common use cases include:
  - Mobile apps
  - Gaming
  - Digital ad serving
  - Live voting
  - Audience interaction for live events
  - Sensor networks
  - Log ingestion
  - Access control for web-based content
  - Metadata storage for Amazon S3 objects
  - E-commerce shopping carts
  - Web session management

- Anti Pattern
  - Prewritten application tied to a traditional relational database: use RDS instead
  - Joins or complex transactions
  - Binary Large Object (BLOB) data: store data in S3 & metadata in DynamoDB
  - Large data with low I/O rate: use S3 instead

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DynamoDB – Read/Write Capacity Modes

- Control how you manage your table's capacity (read/write throughput)

- **Provisioned Mode**
  - You specify the number of reads/writes per second
  - You need to plan capacity beforehand
  - Pay for provisioned read & write capacity units

- **On-Demand Mode (default)**
  - Read/writes automatically scale up/down with your workloads
  - No capacity planning needed
  - Pay for what you use, more expensive ($$$)

- You can switch between different modes once every 24 hours

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R/W Capacity Modes – Provisioned

- Table must have provisioned read and write capacity units
- **Read Capacity Units (RCU) –** throughput for reads
- **Write Capacity Units (WCU) –** throughput for writes
- Option to setup auto-scaling of throughput to meet demand
- Throughput can be exceeded temporarily using **"Burst Capacity"**
- If Burst Capacity has been consumed, you'll get a **"ProvisionedThroughputExceededException"**
- It's then advised to do an **exponential backoff** retry

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DynamoDB – Write Capacity Units (WCU)

- One *Write Capacity Unit (WCU)* represents one write per second for an item up to **1 KB** in size
- If the items are larger than 1 KB, more WCUs are consumed

- **Example 1:** we write 10 items per second, with item size 2 KB
  - We need $10 * (\frac{2\ KB}{1\ KB}) = 20\ WCUs$
- **Example 2:** we write 6 items per second, with item size 4.5 KB
  - We need $6 * (\frac{5\ KB}{1\ KB}) = 30\ WCUs$ (4.5 gets rounded to the upper KB)
- **Example 3:** we write 120 items per minute, with item size 2 KB
  - We need $\left(\frac{120}{60}\right) * (\frac{2\ KB}{1\ KB}) = 4\ WCUs$

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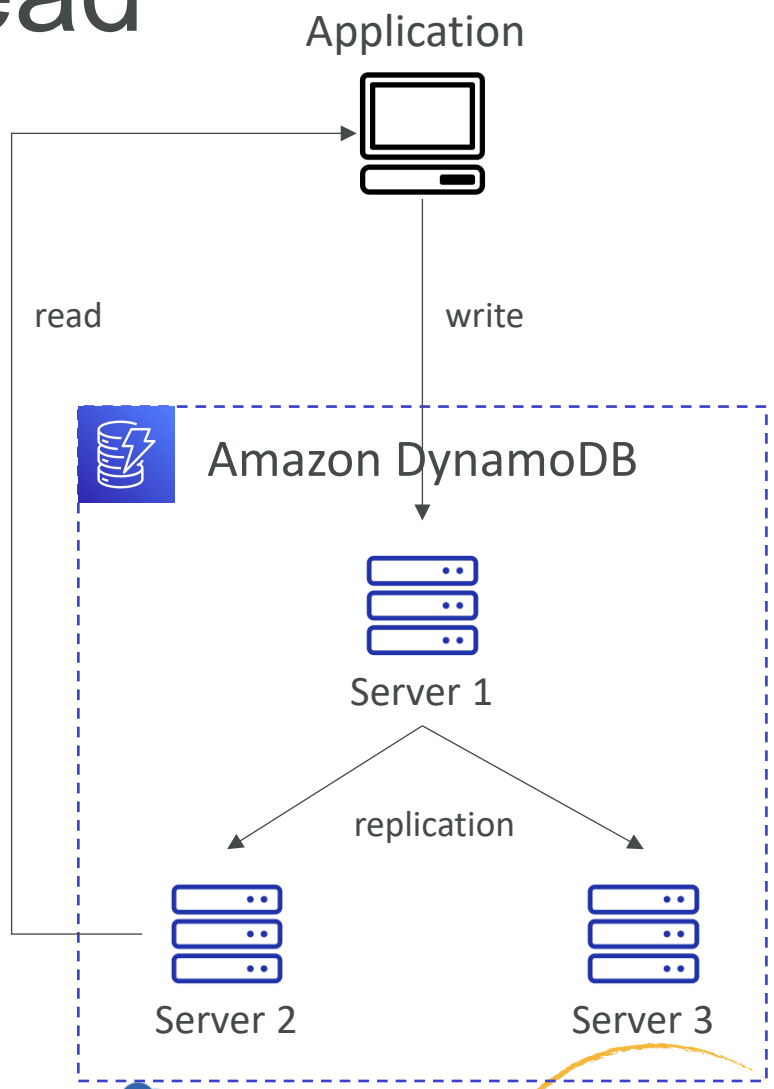
Strongly Consistent Read vs. Eventually Consistent Read

- **Eventually Consistent Read (default)**
  - If we read just after a write, it's possible we'll get some stale data because of replication

- **Strongly Consistent Read**
  - If we read just after a write, we will get the correct data
  - Set **"ConsistentRead"** parameter to **True** in API calls (GetItem, BatchGetItem, Query, Scan)
  - Consumes twice the RCU



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DynamoDB – Read Capacity Units (RCU)

- One *Read Capacity Unit (RCU)* represents **one Strongly Consistent Read** per second, or **two Eventually Consistent Reads** per second, for an item up to **4 KB** in size
- If the items are larger than 4 KB, more RCUs are consumed

- **Example 1:** 10 Strongly Consistent Reads per second, with item size 4 KB
  - We need $10 * \left(\frac{4\ KB}{4\ KB}\right) = 10\ RCUs$
- **Example 2:** 16 Eventually Consistent Reads per second, with item size 12 KB
  - We need $\left(\frac{16}{2}\right) * \left(\frac{12\ KB}{4\ KB}\right) = 24\ RCUs$
- **Example 3:** 10 Strongly Consistent Reads per second, with item size 6 KB
  - We need $10 * \left(\frac{8\ KB}{4\ KB}\right) = 20\ RCUs$ (we must round up 6 KB to 8 KB)

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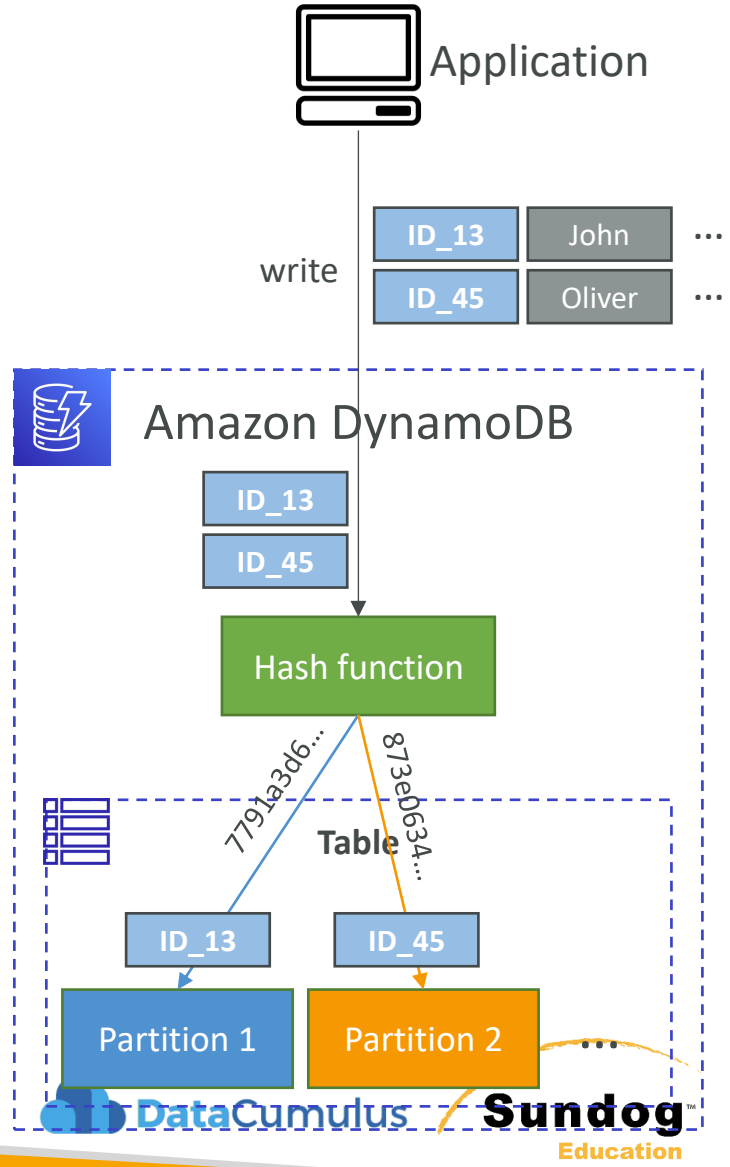
DynamoDB – Partitions Internal

- Data is stored in partitions
- Partition Keys go through a hashing algorithm to know to which partition they go to

- To compute the number of partitions:
  - $\#\ of\ partitions_{by\ capacity} = \left(\frac{RCUs_{Total}}{3000}\right) + \left(\frac{WCUs_{Total}}{1000}\right)$
  - $\#\ of\ partitions_{by\ size} = \frac{Total\ Size}{10\ GB}$
  - $\#\ of\ partitions = \mathrm{ceil}\left(\max\left(\#\ of\ partitions_{by\ capacity}, \#\ of\ partitions_{by\ size}\right)\right)$

- **WCUs and RCUs are spread evenly across partitions**



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DynamoDB – Throttling

- If we exceed provisioned RCUs or WCUs, we get **"ProvisionedThroughputExceededException"**
- Reasons:
  - **Hot Keys** – one partition key is being read too many times (e.g., popular item)
  - **Hot Partitions**
  - **Very large items,** remember RCU and WCU depends on size of items
- Solutions:
  - **Exponential backoff** when exception is encountered (already in SDK)
  - **Distribute partition keys** as much as possible
  - If RCU issue, we can **use DynamoDB Accelerator (DAX)**

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R/W Capacity Modes – On-Demand

- Read/writes automatically scale up/down with your workloads
- No capacity planning needed (WCU / RCU)
- Unlimited WCU & RCU, no throttle, more expensive
- You're charged for reads/writes that you use in terms of **RRU** and **WRU**
- **Read Request Units (RRU) –** throughput for reads (same as RCU)
- **Write Request Units (WRU) –** throughput for writes (same as WCU)
- 2.5x more expensive than provisioned capacity (use with care)
- Use cases: unknown workloads, unpredictable application traffic, …

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DynamoDB – Writing Data

- **PutItem**
  - Creates a new item or fully replace an old item (same Primary Key)
  - Consumes WCUs

- **UpdateItem**
  - Edits an existing item's attributes or adds a new item if it doesn't exist
  - Can be used to implement **Atomic Counters –** a numeric attribute that's unconditionally incremented

- **Conditional Writes**
  - Accept a write/update/delete only if conditions are met, otherwise returns an error
  - Helps with concurrent access to items
  - No performance impact

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DynamoDB – Reading Data

- **GetItem**
  - Read based on Primary key
  - Primary Key can be **HASH** or **HASH+RANGE**
  - Eventually Consistent Read (default)
  - Option to use Strongly Consistent Reads (more RCU - might take longer)
  - **ProjectionExpression** can be specified to retrieve only certain attributes

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DynamoDB – Reading Data (Query)

- **Query** returns items based on:
  - **KeyConditionExpression**
    - Partition Key value (**must be = operator**) – required
    - Sort Key value (=, <, <=, >, >=, Between, Begins with) – optional
  - **FilterExpression**
    - Additional filtering after the Query operation (before data returned to you)
    - Use only with non-key attributes (does not allow HASH or RANGE attributes)
- Returns:
  - The number of items specified in **Limit**
  - Or up to 1 MB of data
- Ability to do pagination on the results
- Can query table, a Local Secondary Index, or a Global Secondary Index

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DynamoDB – Reading Data (Scan)

- **Scan** the entire table and then filter out data (inefficient)
- Returns up to 1 MB of data – use pagination to keep on reading
- Consumes a lot of RCU
- Limit impact using **Limit** or reduce the size of the result and pause
- For faster performance, use **Parallel Scan**
  - Multiple workers scan multiple data segments at the same time
  - Increases the throughput and RCU consumed
  - Limit the impact of parallel scans just like you would for Scans
- Can use **ProjectionExpression** & **FilterExpression** (no changes to RCU)

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DynamoDB – Deleting Data

- **DeleteItem**
  - Delete an individual item
  - Ability to perform a conditional delete

- **DeleteTable**
  - Delete a whole table and all its items
  - Much quicker deletion than calling **DeleteItem** on all items

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DynamoDB – Batch Operations

- Allows you to save in latency by reducing the number of API calls
- Operations are done in parallel for better efficiency
- Part of a batch can fail; in which case we need to try again for the failed items

- **BatchWriteItem**
  - Up to 25 **PutItem** and/or **DeleteItem** in one call
  - Up to 16 MB of data written, up to 400 KB of data per item
  - Can't update items (use **UpdateItem**)
  - **UnprocessedItems** for failed write operations (exponential backoff or add WCU)

- **BatchGetItem**
  - Return items from one or more tables
  - Up to 100 items, up to 16 MB of data
  - Items are retrieved in parallel to minimize latency
  - **UnprocessedKeys** for failed read operations (exponential backoff or add RCU)

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DynamoDB – PartiQL

- SQL-compatible query language for DynamoDB
- Allows you to select, insert, update, and delete data in DynamoDB using SQL
- Run queries across multiple DynamoDB tables
- Run PartiQL queries from:
  - AWS Management Console
  - NoSQL Workbench for DynamoDB
  - DynamoDB APIs
  - AWS CLI
  - AWS SDK

```
SELECT OrderID, Total
FROM Orders
WHERE OrderID IN [1, 2, 3]
ORDER BY OrderID DESC
```

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DynamoDB – Local Secondary Index (LSI)

- **Alternative Sort Key** for your table (same **Partition Key** as that of base table**)**
- The Sort Key consists of one scalar attribute (String, Number, or Binary)
- Up to 5 Local Secondary Indexes per table
- **Must be defined at table creation time**
- **Attribute Projections** – can contain some or all the attributes of the base table (**KEYS_ONLY**, **INCLUDE**, **ALL**)

Primary Key		Attributes		
Partition Key	Sort Key	LSI		
User_ID	**Game_ID**	**Game_TS**	**Score**	**Result**
7791a3d6-...	4421	"2021-03-15T17:43:08"	92	Win
873e0634-...	4521	"2021-06-20T19:02:32"		Lose
a80f73a1-...	1894	"2021-02-11T04:11:31"	77	Win

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DynamoDB – Global Secondary Index (GSI)

- **Alternative Primary Key (HASH or HASH+RANGE)** from the base table
- Speed up queries on non-key attributes
- The Index Key consists of scalar attributes (String, Number, or Binary)
- **Attribute Projections** – some or all the attributes of the base table (**KEYS_ONLY**, **INCLUDE**, **ALL**)
- Must provision RCUs & WCUs for the index
- **Can be added/modified after table creation**

Partition Key	Sort Key	Attributes
User_ID	Game_ID	Game_TS
7791a3d6-...	4421	"2021-03-15T17:43:08"
873e0634-...	4521	"2021-06-20T19:02:32"
a80f73a1-...	1894	"2021-02-11T04:11:31"

TABLE (query by "User_ID")

Partition Key	Sort Key	Attributes
Game_ID	Game_TS	User_ID
4421	"2021-03-15T17:43:08"	7791a3d6-...
4521	"2021-06-20T19:02:32"	873e0634-...
1894	"2021-02-11T04:11:31"	a80f73a1-...

INDEX GSI (query by "Game_ID")

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DynamoDB – Indexes and Throttling

- Global Secondary Index (GSI):
  - **<u>If the writes are throttled on the GSI, then the main table will be throttled!</u>**
  - Even if the WCU on the main tables are fine
  - Choose your GSI partition key carefully!
  - Assign your WCU capacity carefully!

- Local Secondary Index (LSI):
  - Uses the WCUs and RCUs of the main table
  - No special throttling considerations

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DynamoDB - PartiQL

- Use a SQL-like syntax to manipulate DynamoDB tables

Query 1

```
1 SELECT * FROM "demo_indexes" WHERE "user_id" = 'partitionKeyValue' AND
  "game_ts" = 'sortKeyValue'
```

- Supports some (but not all) statements:
  - INSERT
  - UPDATE
  - SELECT
  - DELETE
- It supports Batch operations

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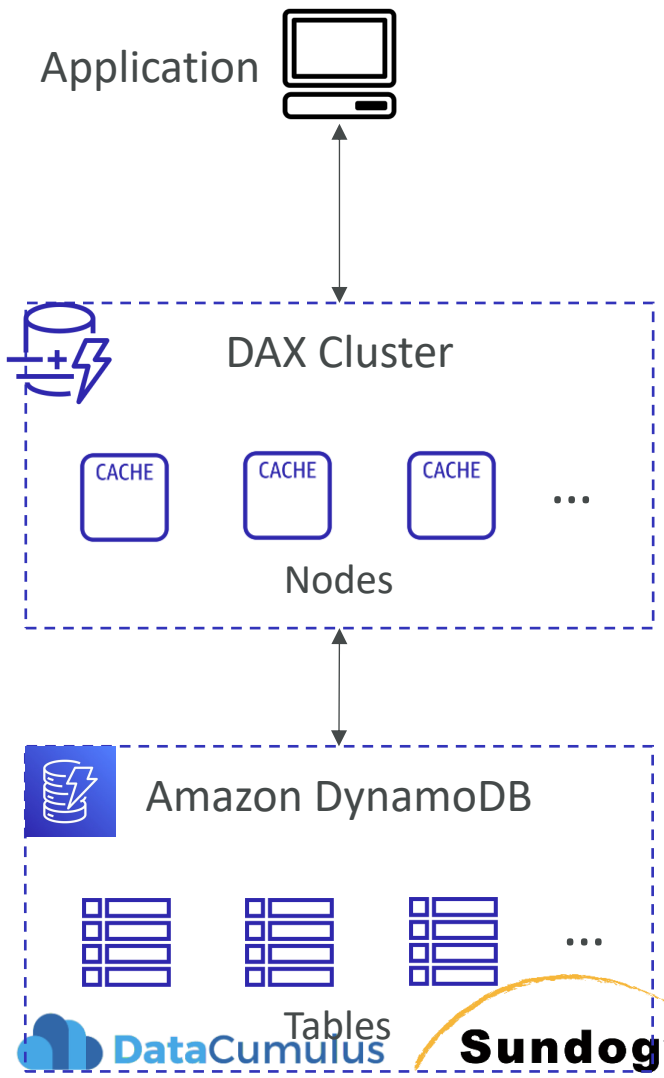
DynamoDB Accelerator (DAX)

- Fully-managed, highly available, seamless in-memory cache for DynamoDB
- Microseconds latency for cached reads & queries
- Doesn't require application logic modification (compatible with existing DynamoDB APIs)
- Solves the "Hot Key" problem (too many reads)
- 5 minutes TTL for cache (default)
- Up to 10 nodes in the cluster
- Multi-AZ (3 nodes minimum recommended for production)
- Secure (Encryption at rest with KMS, VPC, IAM, CloudTrail, …)



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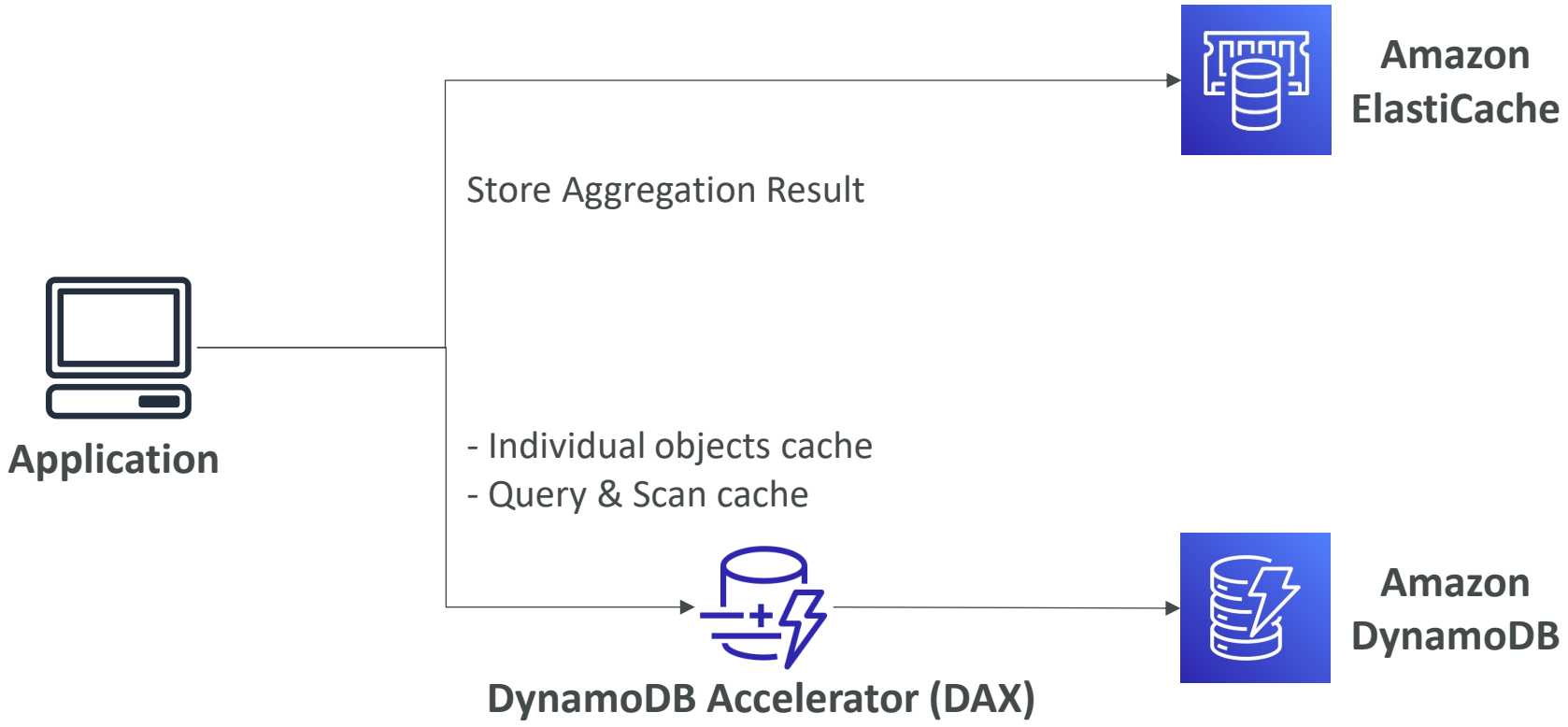
DynamoDB Accelerator (DAX) vs. ElastiCache



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DynamoDB Streams

- Ordered stream of item-level modifications (create/update/delete) in a table
- Stream records can be:
  - Sent to **Kinesis Data Streams**
  - Read by **AWS Lambda**
  - Read by **Kinesis Client Library applications**
- Data Retention for up to 24 hours
- Use cases:
  - react to changes in real-time (welcome email to users)
  - Analytics
  - Insert into derivative tables
  - Insert into OpenSearch Service
  - Implement cross-region replication

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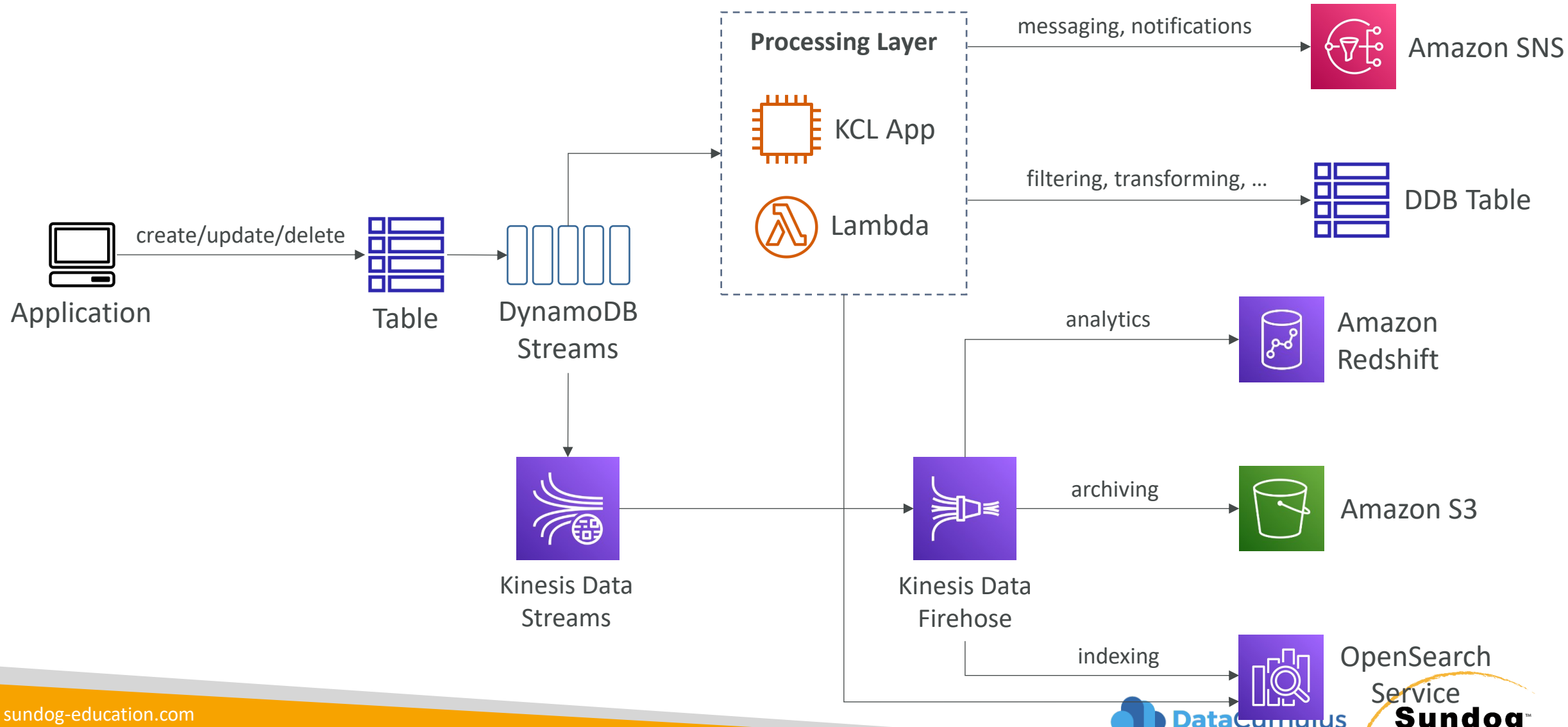
DynamoDB Streams



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DynamoDB Streams

- Ability to choose the information that will be written to the stream:
  - **KEYS_ONLY** – only the key attributes of the modified item
  - **NEW_IMAGE** – the entire item, as it appears after it was modified
  - **OLD_IMAGE** – the entire item, as it appeared before it was modified
  - **NEW_AND_OLD_IMAGES** – both the new and the old images of the item
- DynamoDB Streams are made of shards, just like Kinesis Data Streams
- You don't provision shards, this is automated by AWS

- **Records are not retroactively populated in a stream after enabling it**

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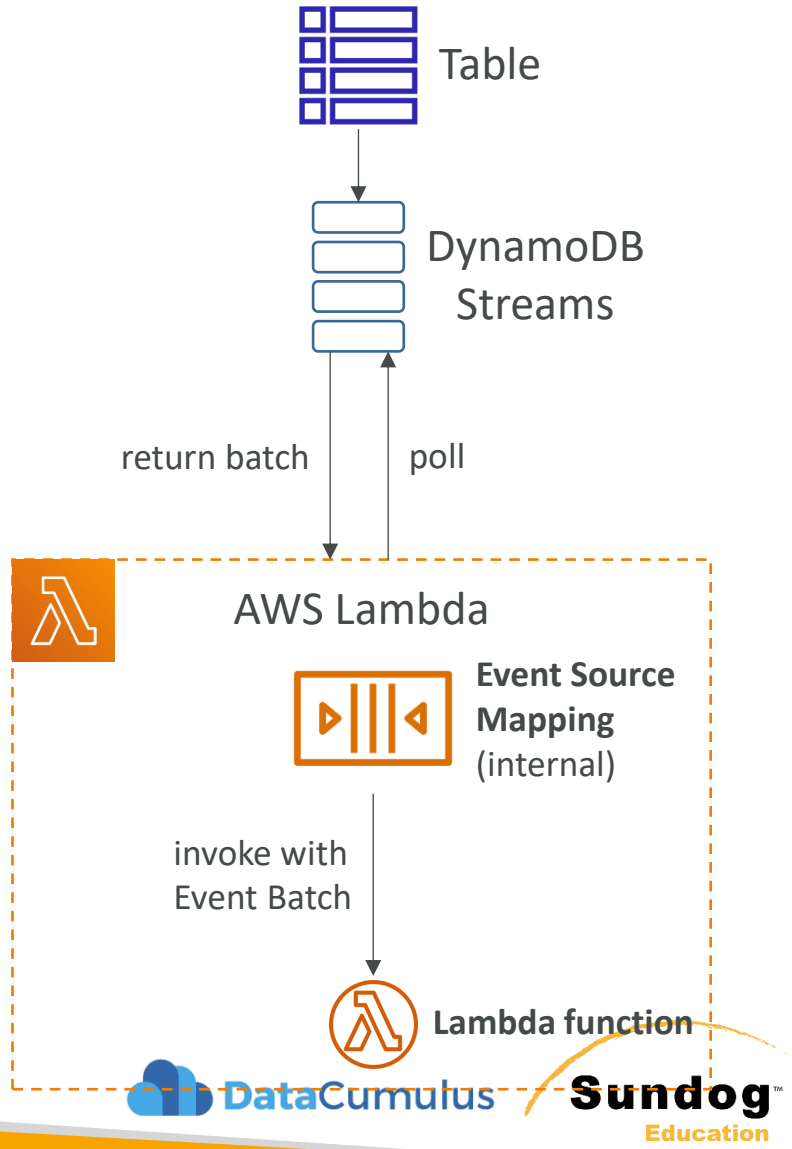
DynamoDB Streams & AWS Lambda

- You need to define an **Event Source Mapping** to read from a DynamoDB Streams
- You need to ensure the Lambda function has the **appropriate permissions**
- **Your Lambda function is invoked synchronously**



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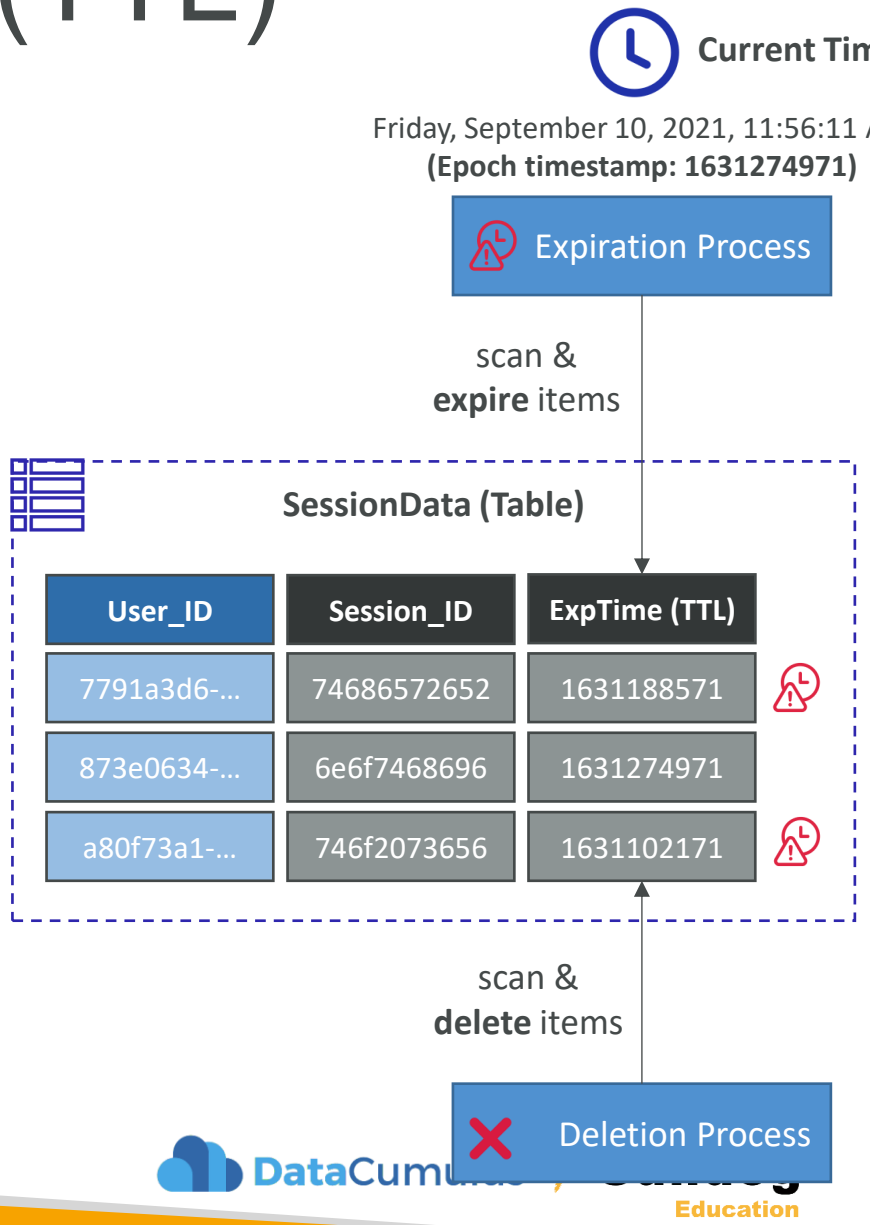
DynamoDB – Time To Live (TTL)

- Automatically delete items after an expiry timestamp
- Doesn't consume any WCUs (i.e., no extra cost)
- The TTL attribute must be a **"Number"** data type with **"Unix Epoch timestamp"** value
- Expired items deleted within few days of expiration
- Expired items, that haven't been deleted, appears in reads/queries/scans (if you don't want them, filter them out)
- Expired items are deleted from both LSIs and GSIs
- A delete operation for each expired item enters the DynamoDB Streams (can help recover expired items)
- Use cases: reduce stored data by keeping only current items, adhere to regulatory obligations, …



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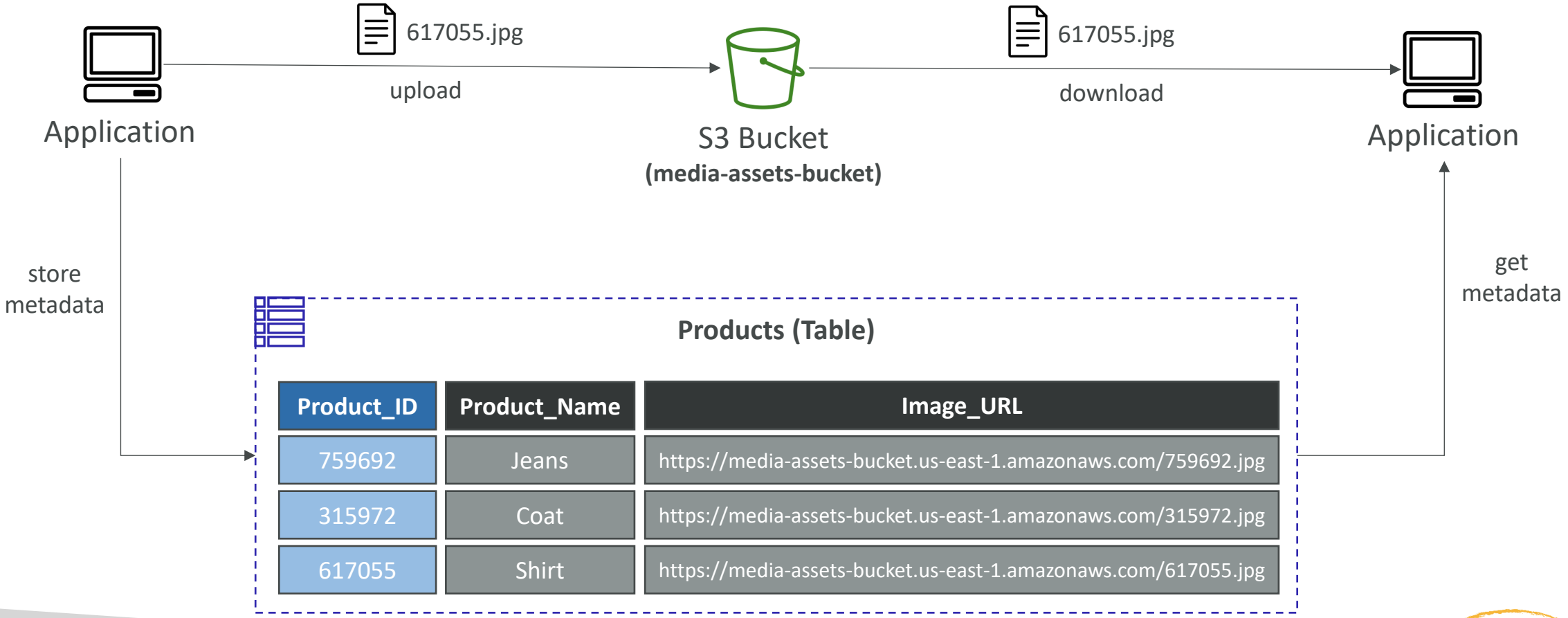
DynamoDB – Large Objects Pattern



Product_ID	Product_Name	Image_URL
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315972	Coat	https://media-assets-bucket.us-east-1.amazonaws.com/315972.jpg
617055	Shirt	https://media-assets-bucket.us-east-1.amazonaws.com/617055.jpg

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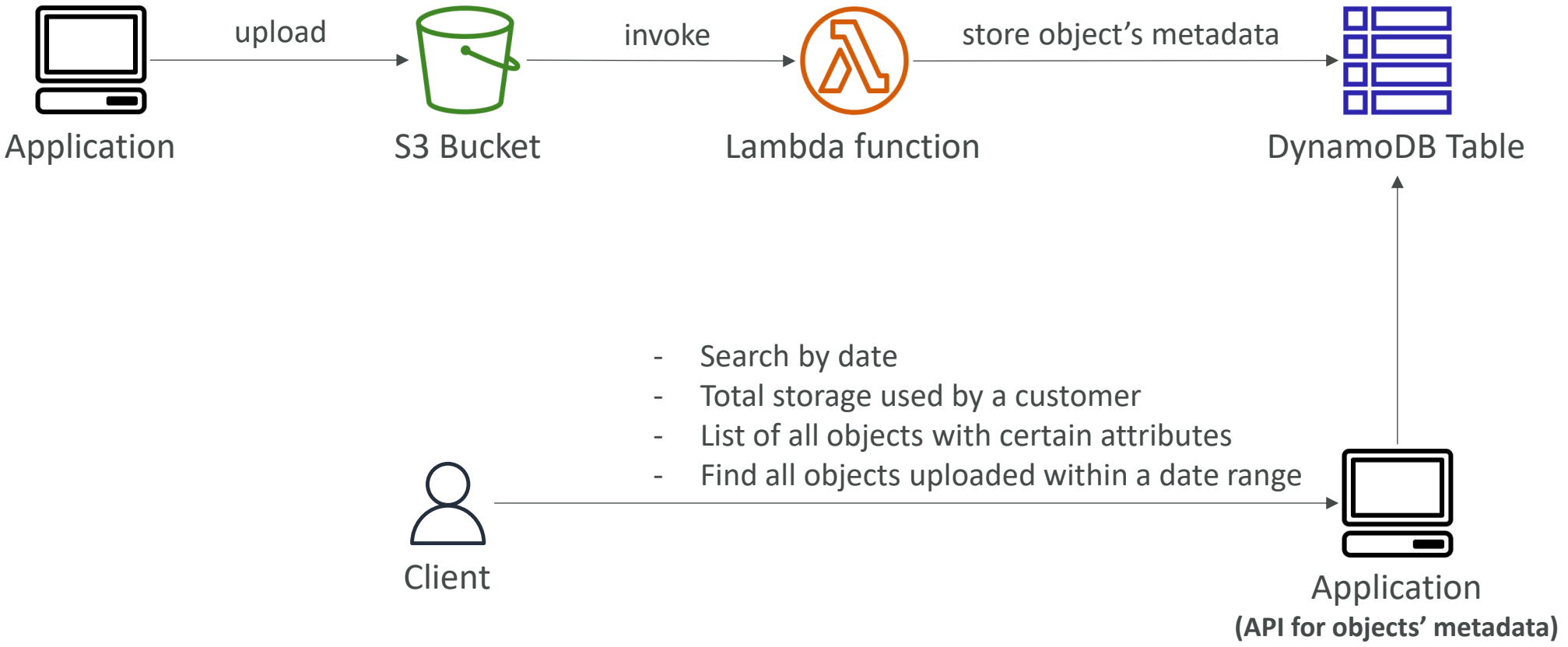
DynamoDB – Indexing S3 Objects Metadata



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DynamoDB – Security & Other Features

- **Security**
  - VPC Endpoints available to access DynamoDB without using the Internet
  - Access fully controlled by IAM
  - Encryption at rest using AWS KMS and in-transit using SSL/TLS
- **Backup and Restore feature available**
  - Point-in-time Recovery (PITR) like RDS
  - No performance impact
- **Global Tables**
  - Multi-region, multi-active, fully replicated, high performance
- **DynamoDB Local**
  - Develop and test apps locally without accessing the DynamoDB web service (without Internet)
- AWS Database Migration Service (AWS DMS) can be used to migrate to DynamoDB (from MongoDB, Oracle, MySQL, S3, …)

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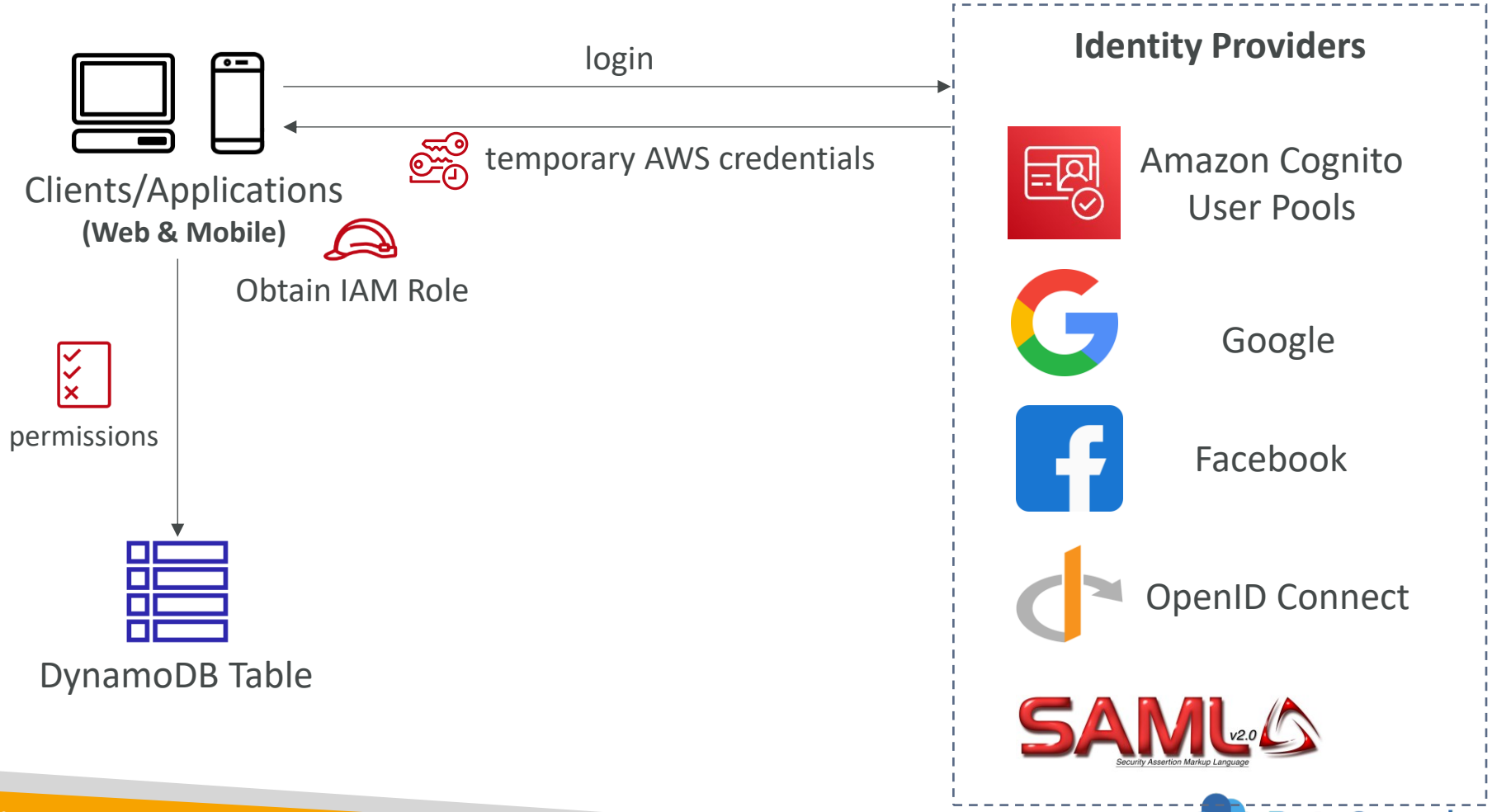
DynamoDB – Users Interact with DynamoDB Directly



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DynamoDB – Fine-Grained Access Control

- Using **Web Identity Federation** or **Cognito Identity Pools**, each user gets AWS credentials
- You can assign an IAM Role to these users with a **Condition** to limit their API access to DynamoDB
- **LeadingKeys –** limit row-level access for users on the Primary Key
- **Attributes –** limit specific attributes the user can see

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "dynamodb:GetIem", "dynamodb:BatchGetItem", "dynamodb:Query",
                "dynamodb:PutItem", "dynamodb:UpdateItem", "dynamodb:DeleteItem",
                "dynamodb:BatchWriteItem"
            ],
            "Resource": "arn:aws:dynamodb:us-west-2:123456789012:table/MyTable",
            "Condition": {
                "ForAllValues:StringEquals": {
                    "dynamodb:LeadingKeys": ["${cognito-identity.amazonaws.com:sub}"]
                }
            }
        }
    ]
}
```

More at: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/specifying-conditions.html

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Amazon RDS

Relational Database Service

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What is RDS?

- Hosted relational database
  - Amazon Aurora
  - MySQL
  - PostgreSQL
  - MariaDB
  - Oracle
  - SQL Server
- Not for “big data”
  - Might appear on exam as an example of what not to use
  - Or in the context of migrating from RDS to Redshift etc.

Amazon RDS

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ACID

- RDS databases offer full ACID compliance
  - Atomicity
  - Consistency
  - Isolation
  - Durability

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Amazon Aurora

- MySQL and PostgreSQL – compatible
- Up to 5X faster than MySQL, 3X faster than PostgreSQL
- 1/10 the cost of commercial databases
- Up to 128TB per database volume
- Up to 15 read replicas
- Continuous backup to S3
- Replication across regions and availability zones
- Automatic scaling with Aurora Serverless

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Aurora Security

- VPC network isolation
- At-rest with KMS
  - Data, backup, snapshots, and replicas can be encrypted
- In-transit with SSL

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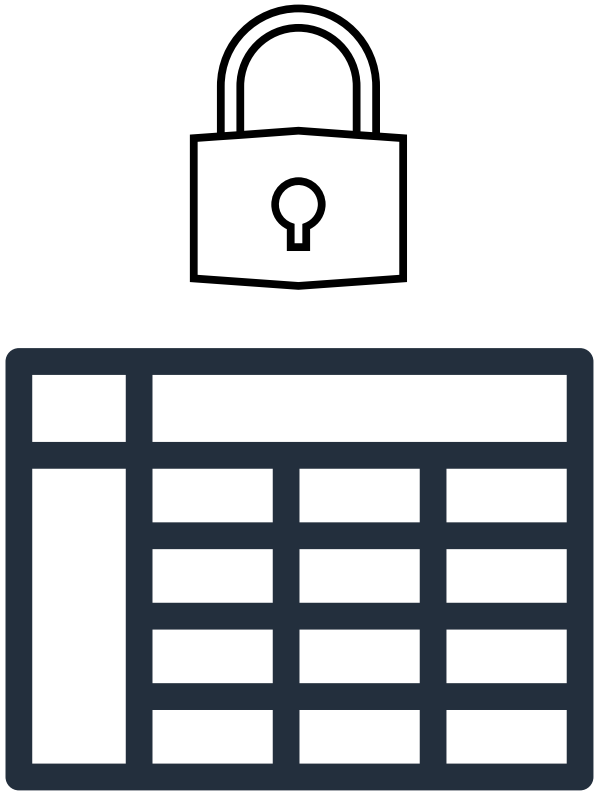
Using the LOCK command

- Relational databases implicitly “lock” tables to prevent two things writing to it at the same time, or reading while a write is in process.
- Tables or rows can also be explicitly locked to ensure data integrity and concurrency control.
- Types of locks:
  - Shared Locks: Allow reads, prevent writes. Can be held by multiple transactions. (FOR SHARE)
  - Exclusive Locks: Prevent all reads and writes to a resource. Only one transaction can hold an exclusive lock. (FOR UPDATE)



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Examples (MySQL)

- Lock an entire table:
  - `LOCK TABLES employees WRITE;` -- Locks the entire 'employees' table for write operations
  - Use `UNLOCK TABLES;` to release the lock.
  - Note: Redshift also has a LOCK command for the same purpose.
- Shared lock (allow reads, prevent other writes during this transaction.)
  - `SELECT * FROM employees WHERE department = 'Finance' FOR SHARE;`
- Exclusive lock (prevent all reads and writes during this transaction)
  - `SELECT * FROM employees WHERE employee_id = 123 FOR UPDATE;`
- Make sure the transactions any locks are in complete, or you could end up with a “deadlock”

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Amazon RDS best practices

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Amazon RDS operational guidelines

- Use CloudWatch to monitor memory, CPU, storage, replica lag
- Perform automatic backups during daily low in write IOPS
- Insufficient I/O will make recovery after failure slow.
  - Migrate to DB instance with more I/O
  - Move to General Purpose or Provisioned IOPS storage
- Set TTL on DNS for your DB instances to 30 seconds or less from your apps
- Test failover before you need it
- Provision enough RAM to include your entire working set
  - If your **ReadIOPS** metric is small and stable, you're good
- Rate limits in Amazon API Gateway can be used to protect your database

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Query optimization in RDS

- Use indexes to accelerate SELECT statements
  - Use EXPLAIN plans to identify the indexes you need
- Avoid full table scans
- Use ANALYZE TABLE periodically
- Simplify WHERE clauses
- Engine-specific optimizations

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DB-specific tweaks

- MySQL, MariaDB
  - Keep tables well under 16TB, ideally under 100GB
  - Have enough RAM to hold indexes of actively used tables
  - Try to have less than 10,000 tables
  - Use InnoDB for storage engine
- PostgreSQL
  - When loading data, disable DB backups and multi-AZ. Tweak various DB parameters such as maintenance_work_mem, max_wal_size, checkpoint_timeout. Disable synchronous_commit, autovacuum, and ensure tables are logged.
  - Use autovacuum

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DB-specific tweaks

- SQL Server
  - Use RDS DB Events to monitor failovers
  - Do not enable simple recover mode, offline mode, or read-only mode (this breaks Multi-AZ)
  - Deploy into all AZ's
- Oracle is its own beast

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DocumentDB

- Aurora is an “AWS-implementation” of PostgreSQL / MySQL …
- **DocumentDB is the same for MongoDB (which is a NoSQL database)**

- MongoDB is used to store, query, and index JSON data
- Similar “deployment concepts” as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB

- Automatically scales to workloads with millions of requests per seconds

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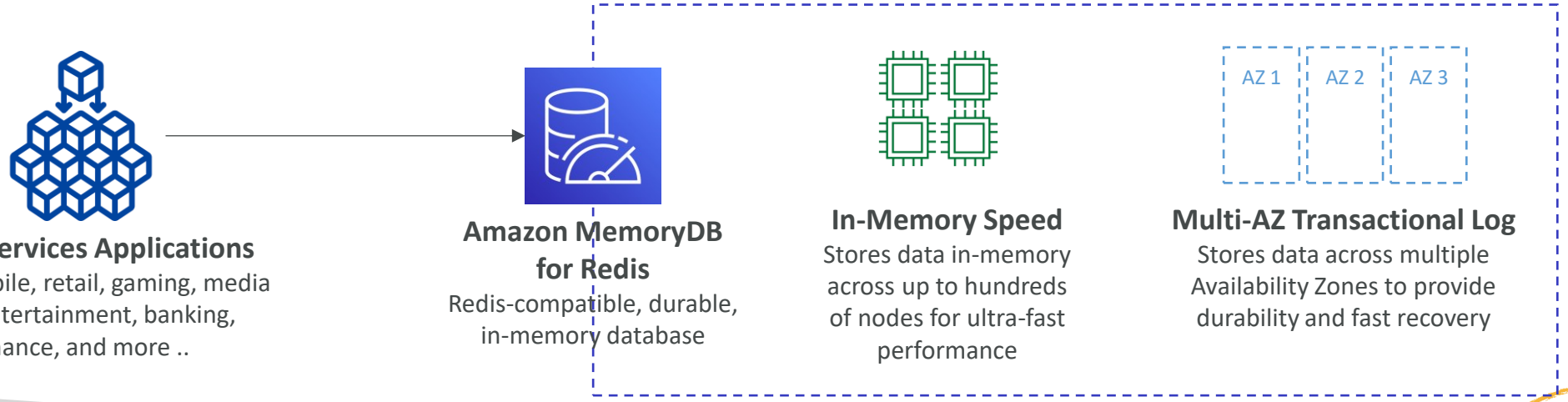
Amazon MemoryDB for Redis

- Redis-compatible, **durable, in-memory database service**
- **Ultra-fast performance with over 160 millions requests/second**
- Durable in-memory data storage with Multi-AZ transactional log
- Scale seamlessly from 10s GBs to 100s TBs of storage
- Use cases: web and mobile apps, online gaming, media streaming, …



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Amazon Keyspaces (for Apache Cassandra)

- *Apache Cassandra is an open-source NoSQL distributed database*

- A managed Apache Cassandra-compatible database service
- Serverless, Scalable, highly available, fully managed by AWS
- Automatically scale tables up/down based on the application's traffic
- Tables are replicated 3 times across multiple AZ
- Using the Cassandra Query Language (CQL)
- Single-digit millisecond latency at any scale, 1000s of requests per second
- Capacity: On-demand mode or provisioned mode with auto-scaling
- Encryption, backup, Point-In-Time Recovery (PITR) up to 35 days

- Use cases: store IoT devices info, time-series data, …

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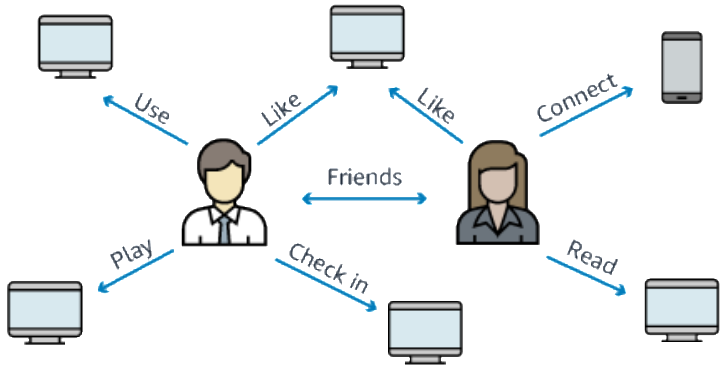
Amazon Neptune

- Fully managed **graph** database
- A popular **graph dataset** would be a **social network**
  - Users have friends
  - Posts have comments
  - Comments have likes from users
  - Users share and like posts…
- Highly available across 3 AZ, with up to 15 read replicas
- Build and run applications working with highly connected datasets – optimized for these complex and hard queries
- Can store up to billions of relations and query the graph with milliseconds latency
- Highly available with replications across multiple AZs
- Great for knowledge graphs (Wikipedia), fraud detection, recommendation engines, social networking



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Amazon Neptune – Query Languges

- Supported graph query languages: Gremlin, openCypher, SPARQL

Neptune

Subject	Predicate	Object
:Alice	:knows	:Bob
:Bob	:knows	:Charlie
:Alice	:age	"30"
:Bob	:age	"25"

SPARQL

```
SELECT ?person
WHERE {
  :Alice :knows ?person .
}
```

Gremlin
$G = (V, E)$

```
g.V().has('name', 'Alice')
      .out('knows').values('name')
```

openCypher

```
MATCH (a:Person {name: "Alice"})-[:KNOWS]->(b)
RETURN b.name
```

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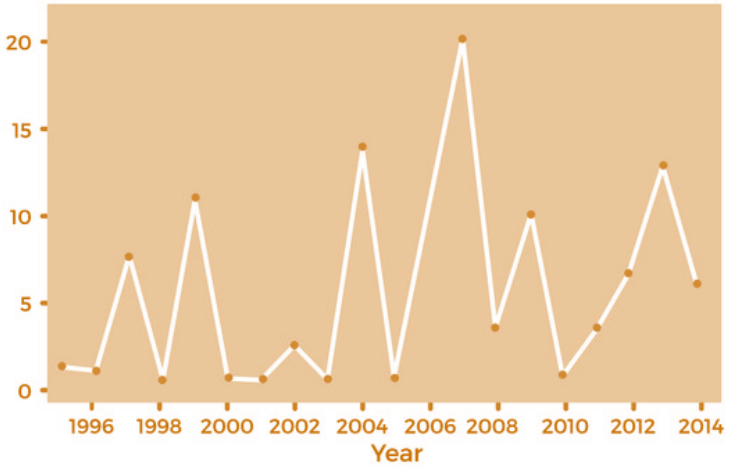
Amazon Timestream

- Fully managed, fast, scalable, serverless **time series database**
- Automatically scales up/down to adjust capacity
- Store and analyze trillions of events per day
- 1000s times faster & 1/10th the cost of relational databases
- Scheduled queries, multi-measure records, SQL compatibility
- Data storage tiering: recent data kept in memory and historical data kept in a cost-optimized storage
- Built-in time series analytics functions (helps you identify patterns in your data in near real-time)
- Encryption in transit and at rest

- Use cases: IoT apps, operational applications, real-time analytics, …



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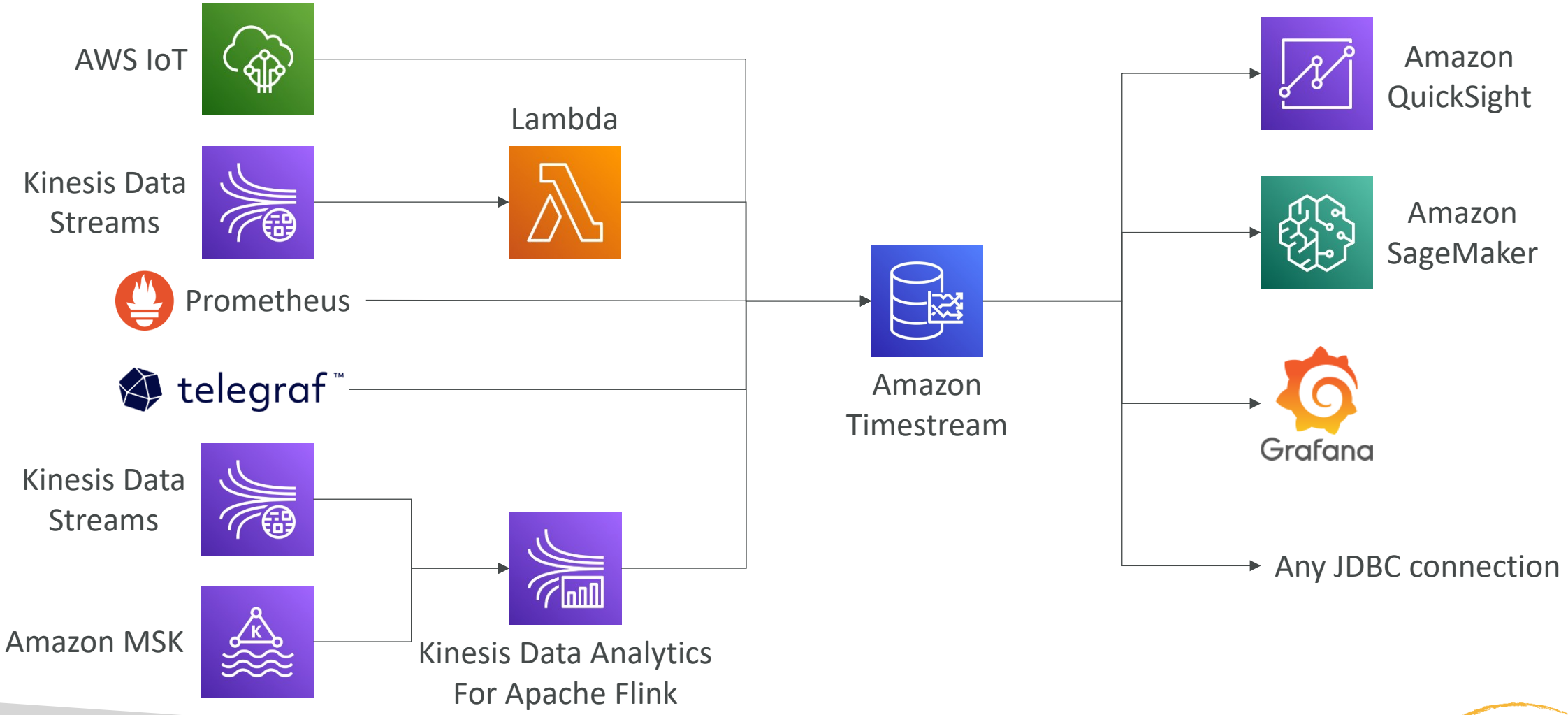
Amazon Timestream – Architecture



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Amazon Redshift

Fully-managed, petabyte-scale data warehouse

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What is Redshift?

- Fully-managed, petabyte scale data warehouse service
- 10X better performance than other DW's
  - Via machine learning, massively parallel query execution, columnar storage
- Designed for OLAP, not OLTP
- Cost effective
- SQL, ODBC, JDBC interfaces
- Scale up or down on demand
- Built-in replication & backups
- Monitoring via CloudWatch / CloudTrail

Amazon Redshift

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Redshift Use-Cases

- Accelerate analytics workloads
- Unified data warehouse & data lake
- Data warehouse modernization
- Analyze global sales data
- Store historical stock trade data
- Analyze ad impressions & clicks
- Aggregate gaming data
- Analyze social trends

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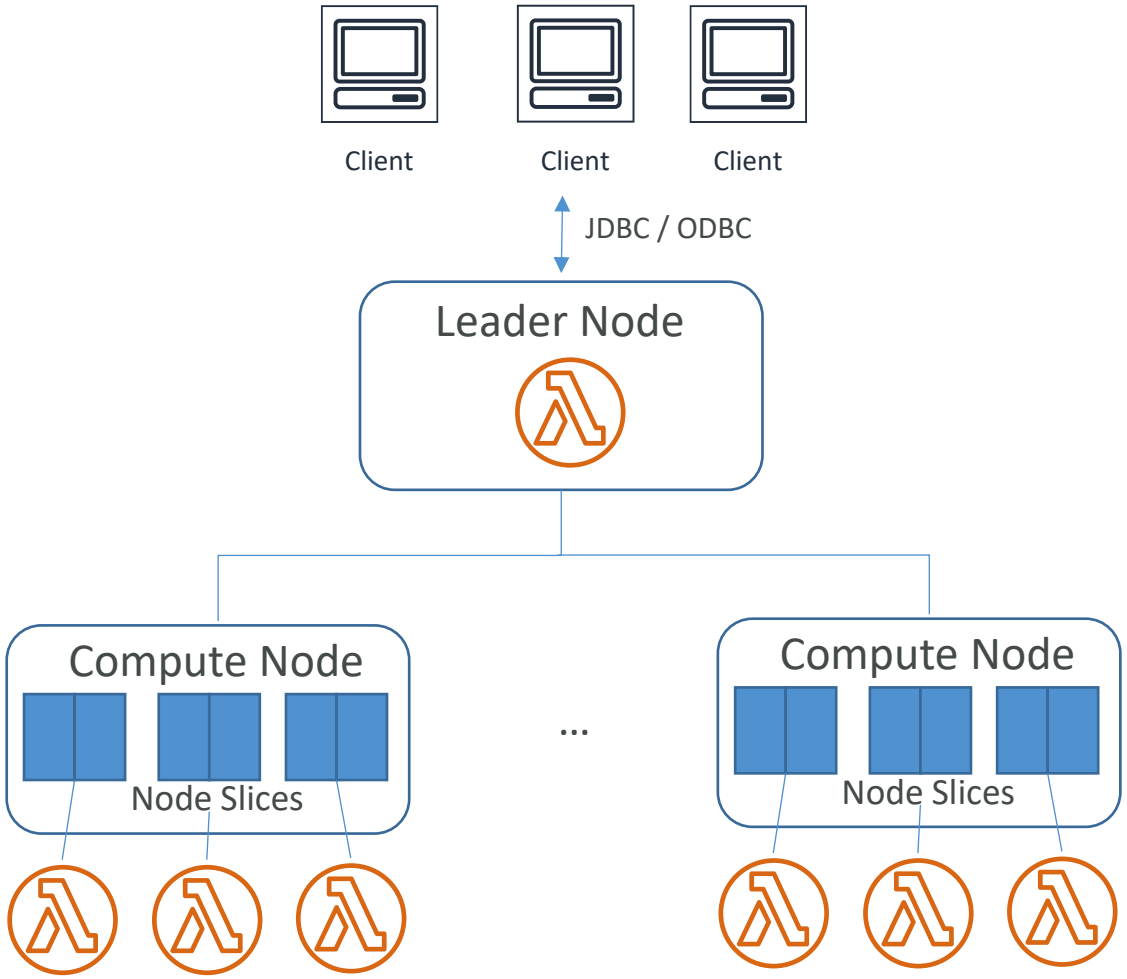
Redshift architecture



Node types:
Dense Storage (DS)
Dense Compute (DC)

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Redshift Spectrum

- Query exabytes of unstructured data in S3 without loading
- Limitless concurrency
- Horizontal scaling
- Separate storage & compute resources
- Wide variety of data formats
- Support of Gzip and Snappy compression

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Redshift Performance

- Massively Parallel Processing (MPP)
- Columnar Data Storage
- Column Compression

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Redshift Durability

- Replication within cluster
- Backup to S3
  - Asynchronously replicated to another region
- Automated snapshots
- Failed drives / nodes automatically replaced
- However – limited to a single availability zone (AZ)
  - Multi-AZ for RA3 clusters now available

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Scaling Redshift

- Vertical and horizontal scaling on demand
- During scaling:
  - A new cluster is created while your old one remains available for reads
  - CNAME is flipped to new cluster (a few minutes of downtime)
  - Data moved in parallel to new compute nodes

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Redshift Distribution Styles

- AUTO
  - Redshift figures it out based on size of data
- EVEN
  - Rows distributed across slices in round-robin
- KEY
  - Rows distributed based on one column
- ALL
  - Entire table is copied to every node

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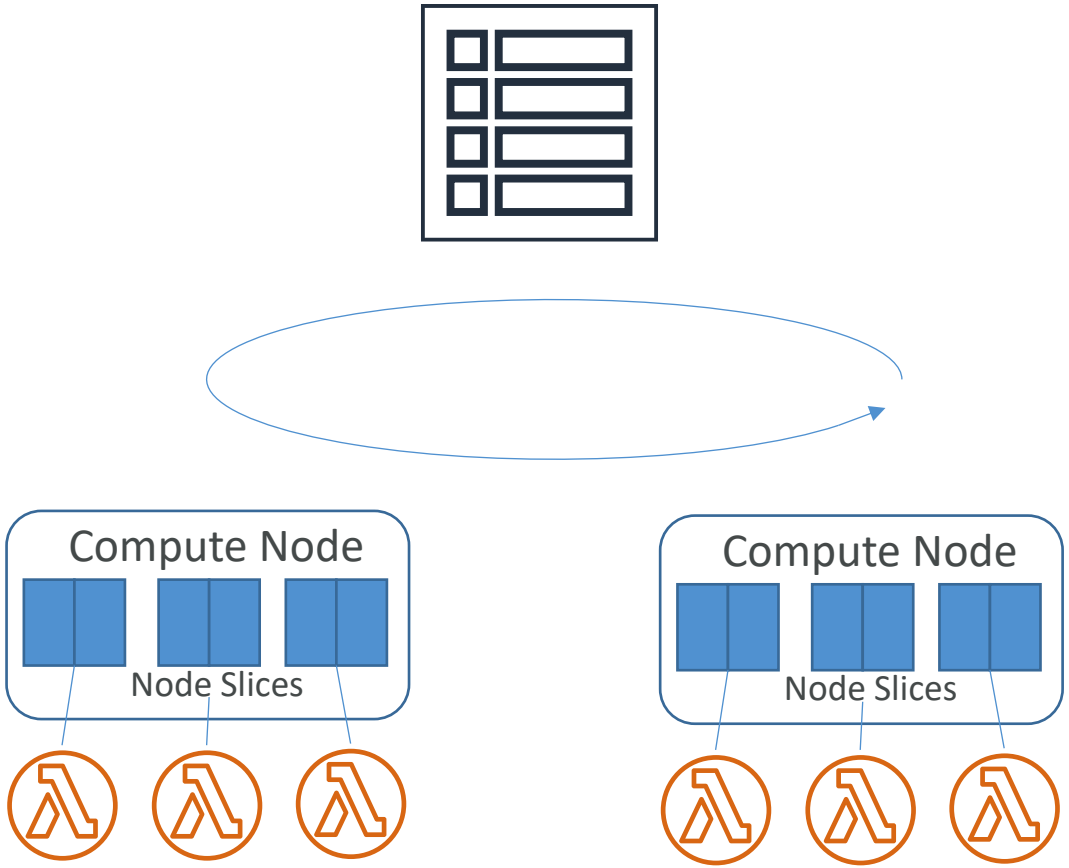
EVEN distribution



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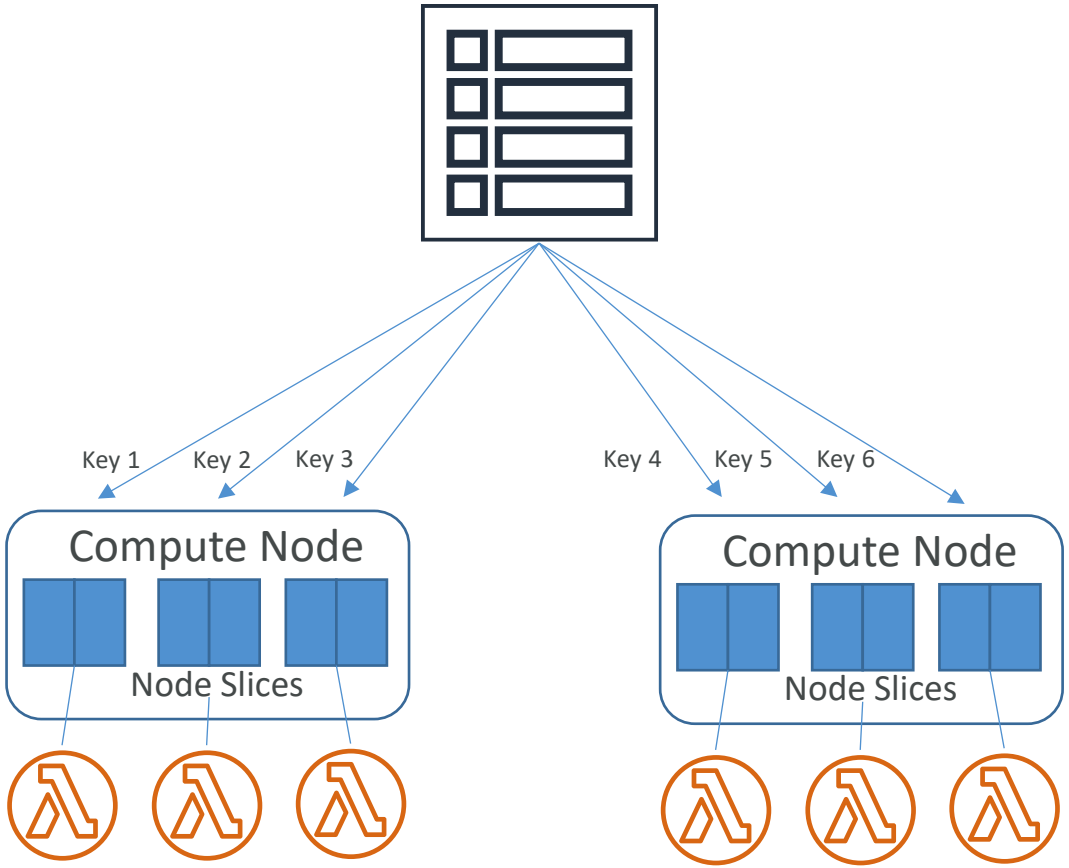
KEY distribution



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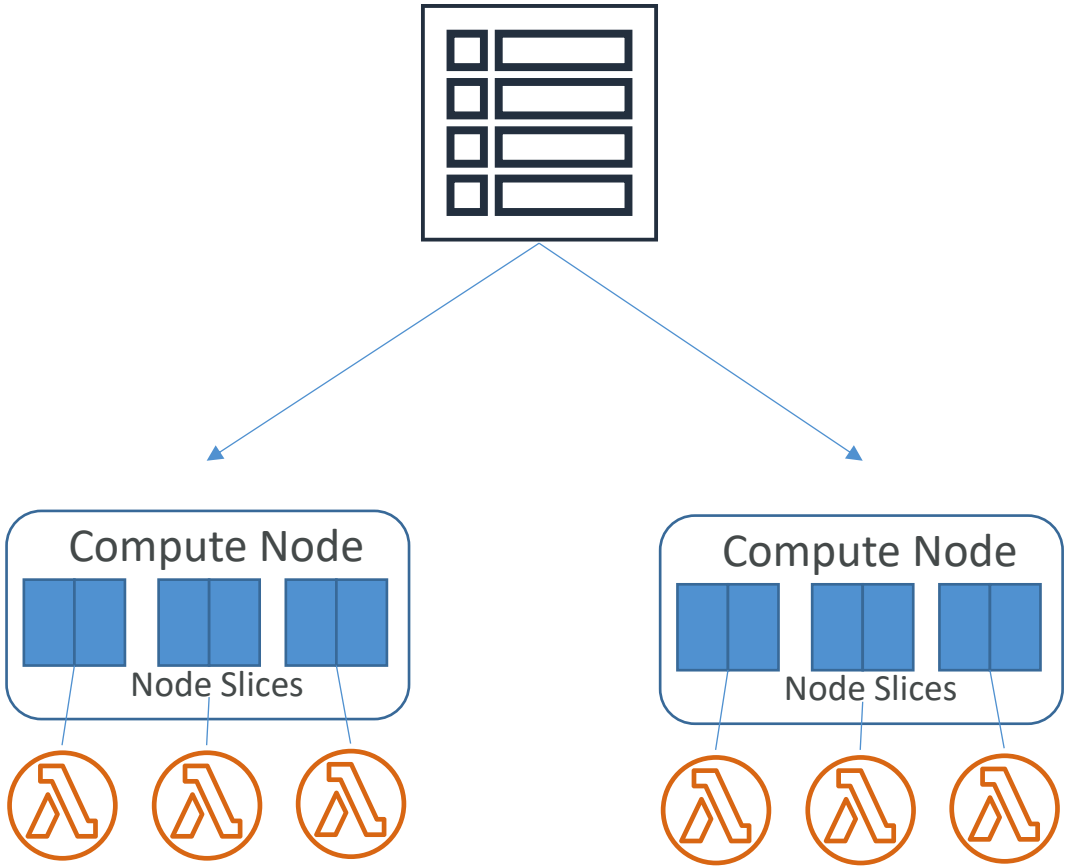
ALL distribution



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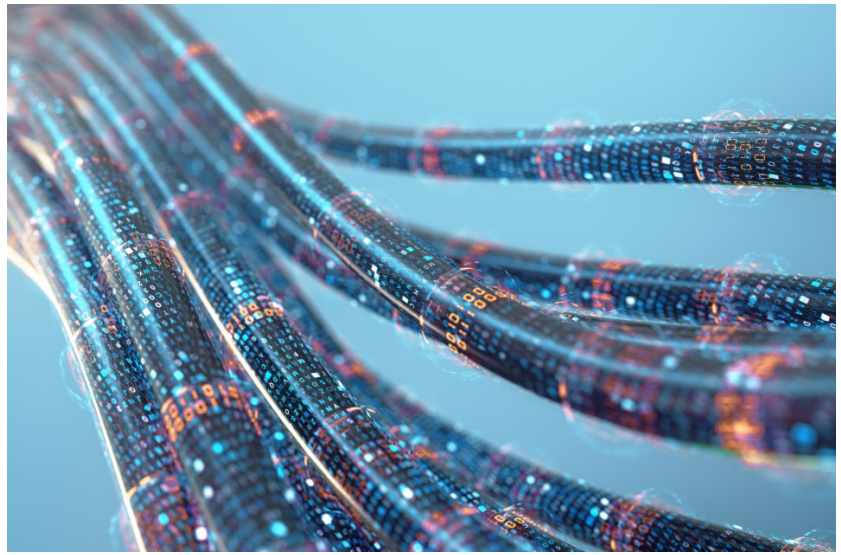
Importing / Exporting data

- COPY command
  - Parallelized; efficient
  - From S3, EMR, DynamoDB, remote hosts
  - S3 requires a manifest file and IAM role
- UNLOAD command
  - Unload from a table into files in S3
- Enhanced VPC routing
- Auto-copy from Amazon S3
- Amazon Aurora zero-ETL integration
  - Auto replication from Aurora -> Redshift
- Redshift Streaming Ingestion
  - From Kinesis Data Streams or MSK



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COPY command: More depth

- Use COPY to load large amounts of data from outside of Redshift
- If your data is already in Redshift in another table,
  - Use INSERT INTO …SELECT
  - Or CREATE TABLE AS
- COPY can decrypt data as it is loaded from S3
  - Hardware-accelerated SSL used to keep it fast
- Gzip, lzop, and bzip2 compression supported to speed it up further
- Automatic compression option
  - Analyzes data being loaded and figures out optimal compression scheme for storing it
- Special case: narrow tables (lots of rows, few columns)
  - Load with a single COPY transaction if possible
  - Otherwise hidden metadata columns consume too much space

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Redshift copy grants for cross-region snapshot copies

- Let's say you have a KMS-encrypted Redshift cluster and a snapshot of it
- You want to copy that snapshot to another region for backup
- In the destination AWS region:
  - Create a KMS key if you don't have one already
  - Specify a unique name for your snapshot copy grant
  - Specify the KMS key ID for which you're creating the copy grant
- In the source AWS region:
  - Enable copying of snapshots to the copy grant you just created

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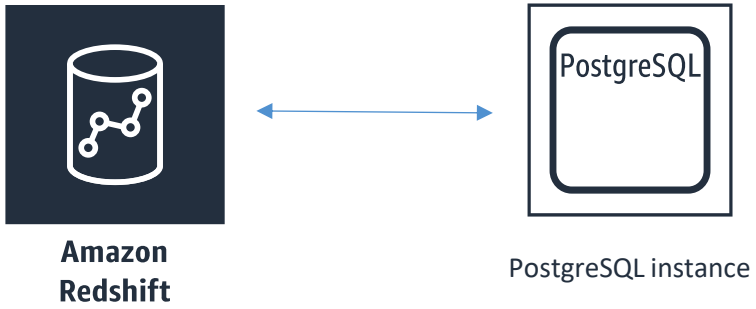
DBLINK

- Connect Redshift to PostgreSQL (possibly in RDS)
- Good way to copy and sync data between PostgreSQL and Redshift




```
CREATE EXTENSION postgres_fdw;
CREATE EXTENSION dblink;
CREATE SERVER foreign_server
    FOREIGN DATA WRAPPER postgres_fdw
    OPTIONS (host '<amazon_redshift _ip>', port '<port>', dbname '<database_name>', sslmode
'require');
CREATE USER MAPPING FOR <rds_postgresql_username>
    SERVER foreign_server
    OPTIONS (user '<amazon_redshift_username>', password '<password>');
```

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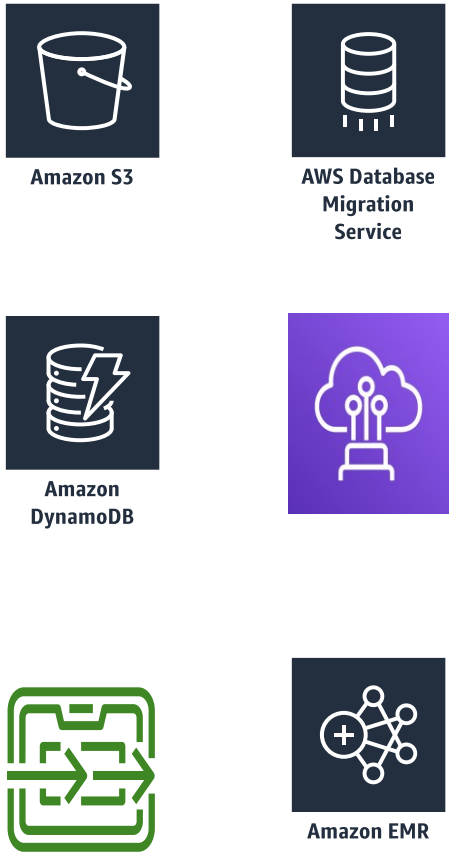
Integration with other services

- S3
- DynamoDB
- EMR / EC2
- Data Pipeline
- Database Migration Service



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Redshift Workload Management (WLM)

- Prioritize short, fast queries vs. long, slow queries
- Query queues
- Via console, CLI, or API

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Concurrency Scaling

- Automatically adds cluster capacity to handle increase in concurrent **read** queries
- Support virtually unlimited concurrent users & queries
- WLM queues manage which queries are sent to the concurrency scaling cluster

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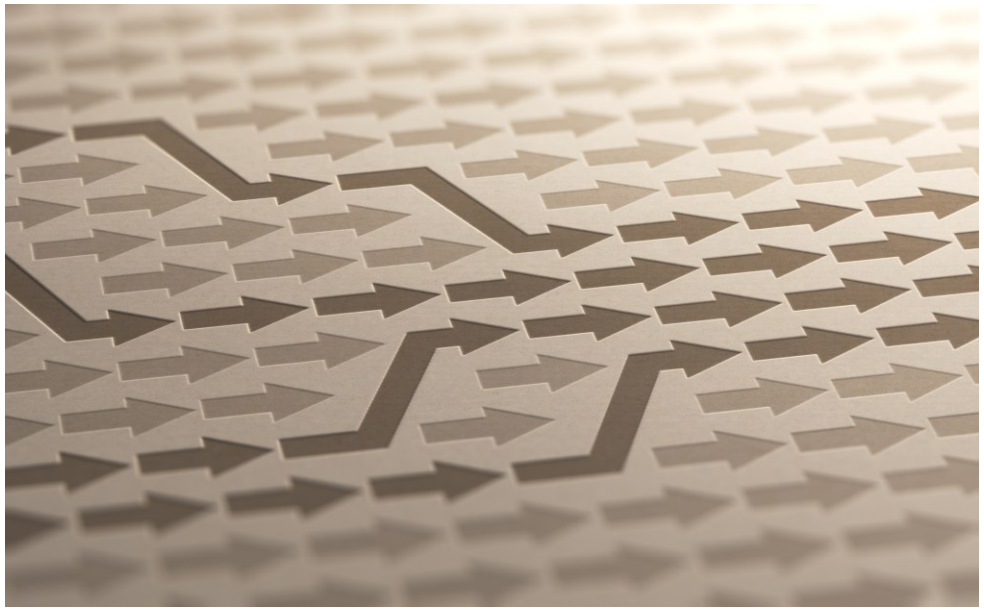
Automatic Workload Management

- Creates up to 8 queues
- Default 5 queues with even memory allocation
- Large queries (ie big hash joins) -> concurrency lowered
- Small queries (ie inserts, scans, aggregations) -> concurrency raised
- Configuring query queues
  - Priority
  - Concurrency scaling mode
  - User groups
  - Query groups
  - Query monitoring rules



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Manual Workload Management

- One default queue with concurrency level of 5 (5 queries at once)
- Superuser queue with concurrency level 1
- Define up to 8 queues, up to concurrency level 50
  - Each can have defined concurrency scaling mode, concurrency level, user groups, query groups, memory, timeout, query monitoring rules
  - Can also enable query queue hopping
    - Timed out queries “hop” to next queue to try again

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Short Query Acceleration (SQA)

- Prioritize short-running queries over longer-running ones
- Short queries run in a dedicated space, won't wait in queue behind long queries
- Can be used in place of WLM queues for short queries
- Works with:
  - CREATE TABLE AS (CTAS)
  - Read-only queries (SELECT statements)
- Uses machine learning to predict a query's execution time
- Can configure how many seconds is "short"

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Resizing Redshift Clusters

- Elastic resize
  - Quickly add or remove nodes of same type
    - (It *can* change node types, but not without dropping connections – it creates a whole new cluster)
  - Cluster is down for a few minutes
  - Tries to keep connections open across the downtime
  - Limited to doubling or halving for some dc2 and ra3 node types.
- Classic resize
  - Change node type and/or number of nodes
  - Cluster is read-only for hours to days
- Snapshot, restore, resize
  - Used to keep cluster available during a classic resize
  - Copy cluster, resize new cluster

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VACUUM command

- Recovers space from deleted rows and restores sort order
- VACUUM FULL
- VACUUM DELETE ONLY
  - Skips the sort
- VACUUM SORT ONLY
  - Does not reclaim space!
- VACUUM REINDEX
  - Re-analyzes distribution of sort key columns
  - Then does a full VACUUM

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Newer Redshift features

- RA3 nodes with managed storage
  - Enable independent scaling of compute and storage
  - SSD-based
- Redshift data lake export
  - Unload Redshift query to S3 in Apache Parquet format
  - Parquet is 2x faster to unload and consumes up to 6X less storage
  - Compatible with Redshift Spectrum, Athena, EMR, SageMaker
  - Automatically partitioned
- Spatial data types
  - GEOMETRY, GEOGRAPHY
- Cross-Region Data Sharing
  - Share live data across Redshift clusters without copying
  - Requires new RA3 node type
  - Secure, across regions and across accounts

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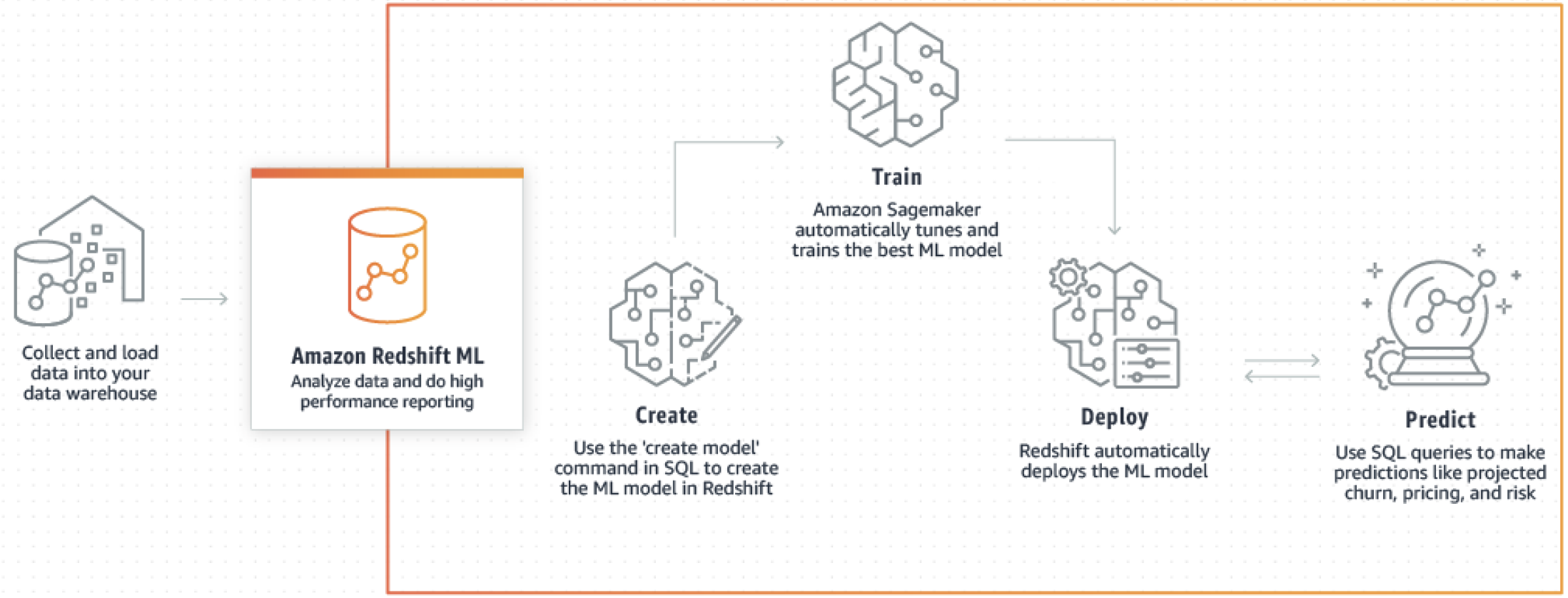
Amazon Redshift ML



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Redshift anti-patterns

- Small data sets
  - Use RDS instead
- OLTP
  - Use RDS or DynamoDB instead
- Unstructured data
  - ETL first with EMR etc.
- BLOB data
  - Store references to large binary files in S3, not the files themselves.

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Redshift security concerns

- Using a Hardware Security Module (HSM)
  - Must use a client and server certificate to configure a trusted connection between Redshift and the HSM
  - If migrating an unencrypted cluster to an HSM-encrypted cluster, you must create the new encrypted cluster and then move data to it.
- Defining access privileges for user or group
  - Use the GRANT or REVOKE commands in SQL
  - Example: grant select on table foo to bob;

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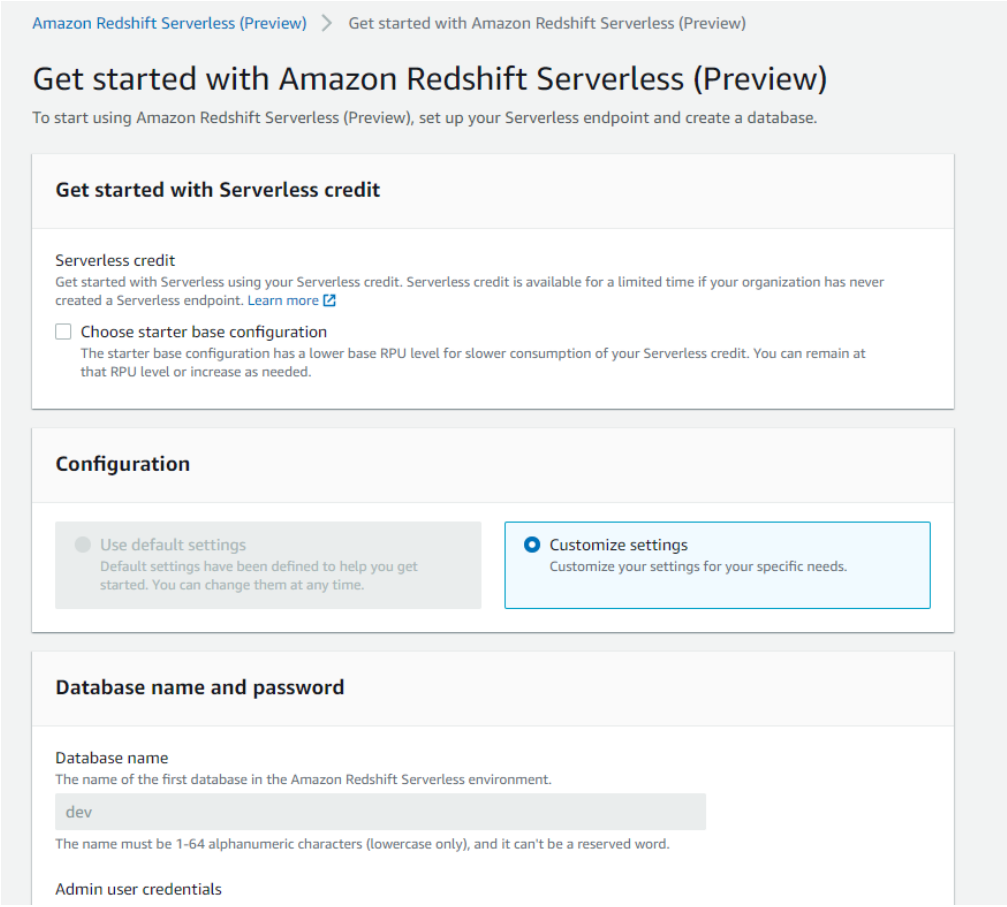
Redshift Serverless

- Automatic scaling and provisioning for your workload
- Optimizes costs & performance
  - Pay only when in use
- Uses ML to maintain performance across variable & sporadic workloads
- Easy spinup of development and test environments
- Easy ad-hoc business analysis
- You get back a serverless endpoint, JDBC/ODBC connection, or just query via the console's query editor.



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Redshift Serverless: Getting Started

- Need an IAM role with this policy
- Define your
  - Database name
  - Admin user credentials
  - VPC
  - Encryption settings
    - AWS-owned KMS by default
  - Audit logging
- Can manage snapshots & recovery points after creation

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "redshift-serverless:*",
      "Resource": "*"
    }
  ]
}
```

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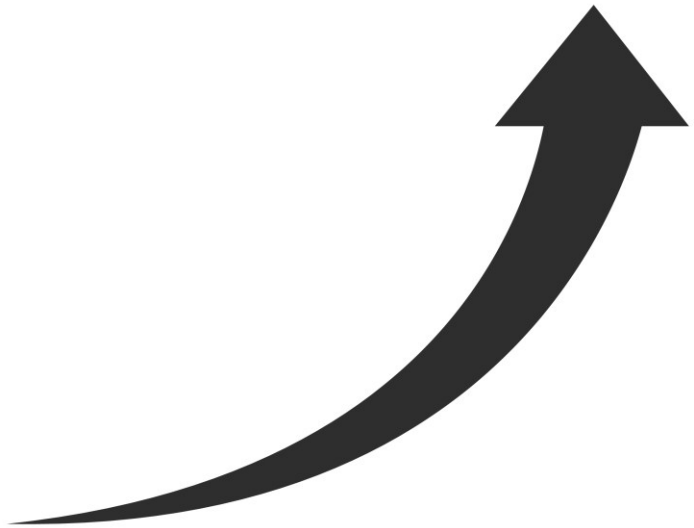
Resource Scaling in Redshift Serverless

- Capacity measured in Redshift Processing Units (RPU's)
- You pay for RPU-hours (per second) plus storage
- Base RPU's
  - You can adjust base capacity
  - Defaults to AUTO
  - But you can adjust from 32-512 RPU's to improve query performance
- Max RPU's
  - Can set a usage limit to control costs
  - Or, increase it to improve throughput



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Redshift Serverless

- Does everything Redshift can, except:
  - Parameter Groups
  - Workload Management
  - AWS Partner integration
  - Maintenance windows / version tracks
- No public endpoints (yet)
  - Must access within a VPC

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Redshift Serverless: Monitoring

- Monitoring views
  - SYS_QUERY_HISTORY
  - SYS_LOAD_HISTORY
  - SYS_SERVERLESS_USAGE
  - …and many more
- CloudWatch logs
  - Connection & user logs enabled by default
  - Optional user activity log data
  - Under /aws/redshift/serverless/
- CloudWatch metrics
  - QueriesCompletedPerSecond, QueryDuration, QueriesRunning,etc.
  - Dimensions: DatabaseName, latency (short/medium/long), QueryType, stage

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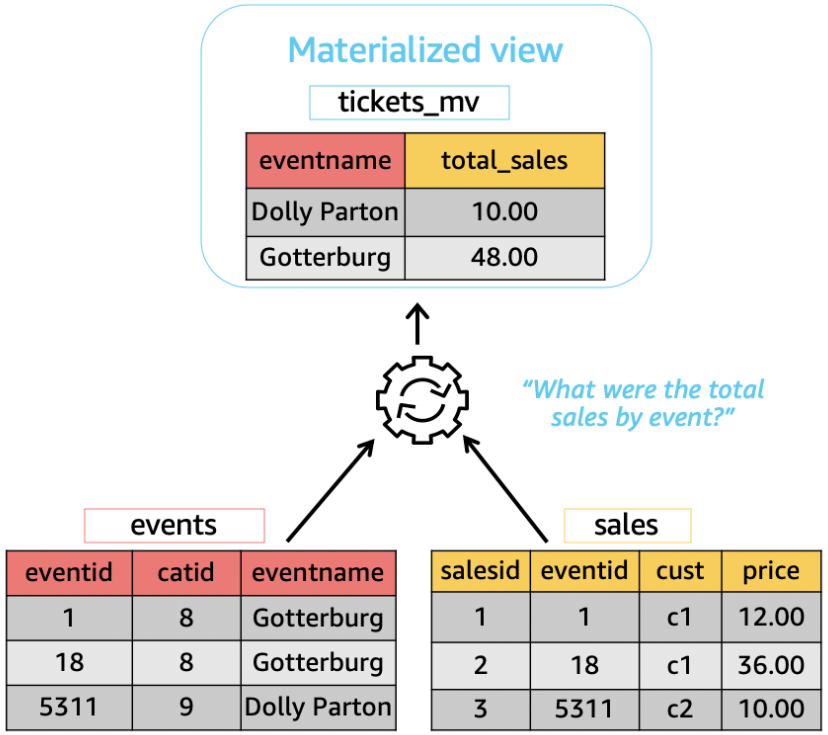
Redshift Materialized Views

- Contain precomputed results based on SQL queries over one or more base tables.
  - This differs from a “normal” view in that it actually stores the results of the query
- Provide a way to speed up complex queries in a data warehouse environment, especially on large tables.
- You can query materialized views just like any other tables or views.
- Queries return results faster since they use precomputed results without accessing base tables.
- They're particularly beneficial for predictable and recurring queries, e.g., populating dashboards like Amazon QuickSight.



AWS

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Using Materialized Views

- CREATE MATERIALIZED VIEW…
- Keeping them refreshed
  - REFRESH MATERIALIZED VIEW…
  - Set AUTO REFRESH option on creation
- Query them just like any other table or view
- Materialized views can be built from other materialized views
  - Useful for re-using expensive joins

```
CREATE MATERIALIZED VIEW tickets_mv AS
    select    catgroup,
    sum(qtysold) as sold
    from      category c, event e, sales s
    where     c.catid = e.catid
    and       e.eventid = s.eventid
    group by catgroup;
```

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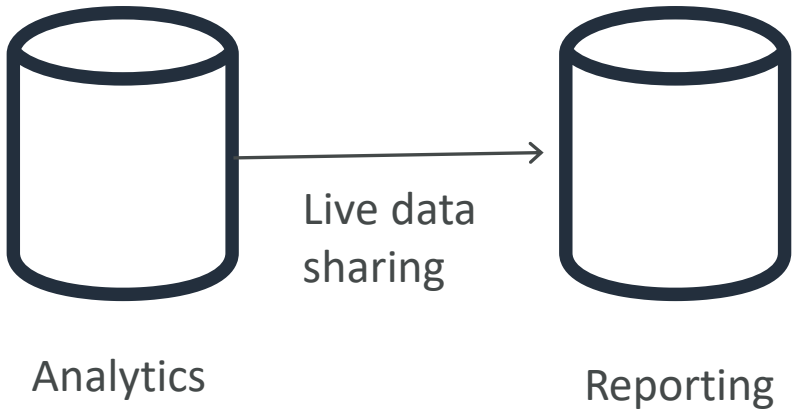
Redshift Data Sharing

- Securely share live data across Redshift clusters for read purposes
- Why?
  - Workload isolation
  - Cross-group collaboration
  - Sharing data between dev/test/prod
  - Licensing data access in AWS Data Exchange
- Can share DB's, schemas, tables, views, and/or UDFs.
  - Fine-grained access control



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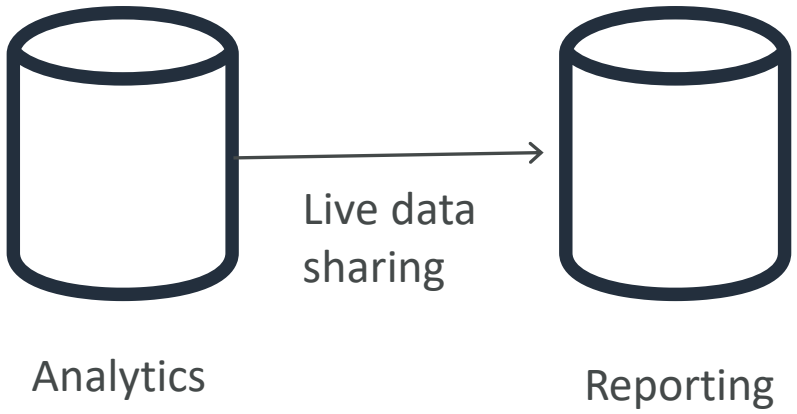
Redshift Data Sharing

- Producer / consumer architecture
  - Producer controls security
  - Isolation to ensure producer performance unaffected by consumers
  - Data is live and transactionally consistent
- Both must be encrypted, must use RA3 nodes
- Cross-region data sharing involves transfer charges
- Types of data shares
  - Standard
  - AWS Data Exchange
  - AWS Lake Formation - managed



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Redshift Lambda UDF

- Use custom functions in AWS Lambda inside SQL queries
  - Using any language you want!
  - Do anything you want!
    - Call other services (AI?)
    - Access external systems
    - Integrate with location service
- Register with CREATE EXTERNAL FUNCTION
- Must GRANT USAGE ON LANGUAGE EXFUNC for permissions

```
SELECT a, b FROM t1 WHERE
lambda_multiply(a, b) = 64;
```

```
CREATE EXTERNAL FUNCTION exfunc_sum(INT,INT) RETURNS
INT VOLATILE LAMBDA 'lambda_sum' IAM_ROLE
'arn:aws:iam::123456789012:role/Redshift-Exfunc-Test';
```

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Redshift Lambda UDF

- Use AWSLambdaRole IAM policy to grant permissions to Lambda on your cluster's IAM role
  - Or roll your own policy, to allow lambda:InvokeFunction
- You also need this IAM role in your CREATE EXTERNAL FUNCTION command
  - Possible to invoke functions in other accounts in the same Region, using IAM role chaining
- Redshift communicates with Lambda using JSON


```
{
 "request_id" : "23FF1F97-F28A-44AA-AB67-266ED976BF40",
 "cluster" : "arn:aws:redshift:xxxx",
 "user" : "adminuser",
 "database" : "db1",
 "external_function": "public.foo",
 "query_id" : 5678234,
 "num_records" : 4,
 "arguments" : [
   [ 1, 2 ],
   [ 3, null],
   null,
   [ 4, 6]
  ]
 }
```



```
{
 "success": true,   // true indicates the call succeeded
 "error_msg" : "my function isn't working",  // shall only exist when success != true
 "num_records": 4,     // number of records in this payload
 "results" : [
   1,
   4,
   null,
   7
  ]
}
```


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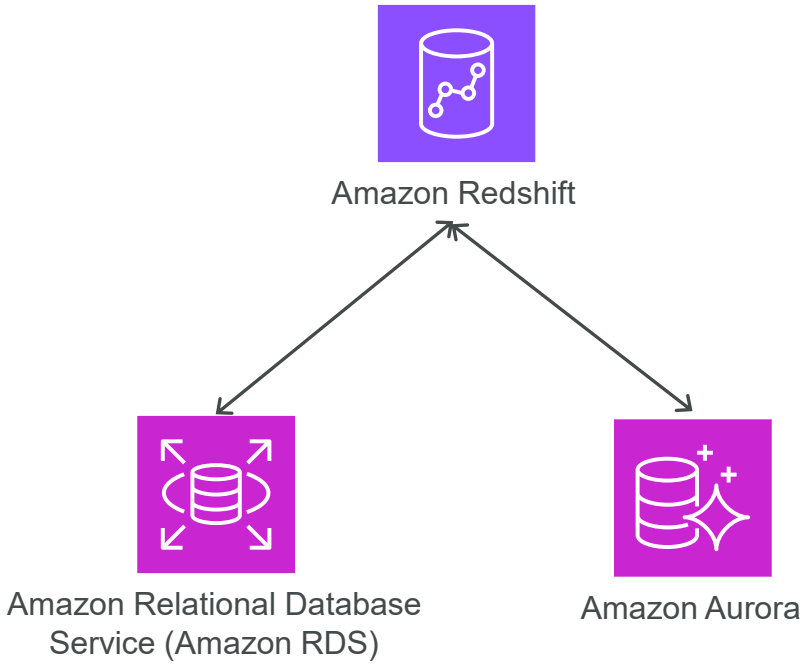
Redshift Federated Queries

- Query and analyze across databases, warehouses, and lakes
- Ties Redshift to Amazon RDS or Aurora for PostgreSQL and MySQL
  - Incorporate live data in RDS into your Redshift queries
  - Avoids the need for ETL pipelines
- Offloads computation to remote databases to reduce data movement



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Redshift Federated Queries

- Must establish connectivity between your Redshift cluster and RDS / Aurora
  - Put them in the same VPC subnet
  - Or use VPC peering
- Credentials must be in AWS Secrets Manager
- Include secrets in IAM role for your Redshift cluster
- Connect using CREATE EXTERNAL SCHEMA
  - Can also connect to S3 / Redshift Spectrum this way
- The SVV_EXTERNAL_SCHEMAS view contains available external schemas


```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AccessSecret",
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetResourcePolicy",
        "secretsmanager:GetSecretValue",
        "secretsmanager:DescribeSecret",
        "secretsmanager:ListSecretVersionIds"
      ],
      "Resource": "arn:aws:secretsmanager:us-west-
2:123456789012:secret:my-rds-secret-VNenFy"
    },
    {
      "Sid": "VisualEditor1",
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetRandomPassword",
        "secretsmanager:ListSecrets"
      ],
      "Resource": "*"
    }
  ]
}
```


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Redshift Federated Queries

- Read-only access to external data sources
- Costs will be incurred on external DB's
- You can query RDS/Aurora from Redshift, but not the other way around

```
CREATE EXTERNAL SCHEMA apg
FROM POSTGRES
DATABASE 'database-1' SCHEMA 'myschema'
URI 'endpoint to aurora hostname'
IAM_ROLE 'arn:aws:iam::123456789012:role/Redshift-SecretsManager-RO'
SECRET_ARN 'arn:aws:secretsmanager:us-west-2:123456789012:secret:federation/test/dataplane-apg-creds-YbVKQw';
```

```
SELECT count(*) FROM apg.lineitem;

count
-------
11760
```

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Redshift System Tables and Views

- Contains info about how Redshift is functioning
- Types of system tables / views
  - SYS views: Monitor query & workload usage
  - STV tables: Transient data containing snapshots of current system data
  - SVV views: metadata about DB objects that reference STV tables
  - STL views: Generated from logs persisted to disk
  - SVCS views: Details about queries on main & concurrency scaling clusters
  - SVL views: Details about queries on main clusters
- Many system monitoring views & tables are only for provisioned clusters, not serverless

```
SELECT q.query, q.starttime, q.endtime,
       datediff(seconds, q.starttime, q.endtime) as
execution_time,
       q.aborted, qr.queue_time, qr.exec_time,
qr.total_exec_time
FROM stl_query q
JOIN svl_qlog qr ON q.query = qr.query
WHERE q.starttime > (GETDATE() - INTERVAL '1 day')
ORDER BY q.starttime DESC;
```

Example: analyze execution time of recent queries

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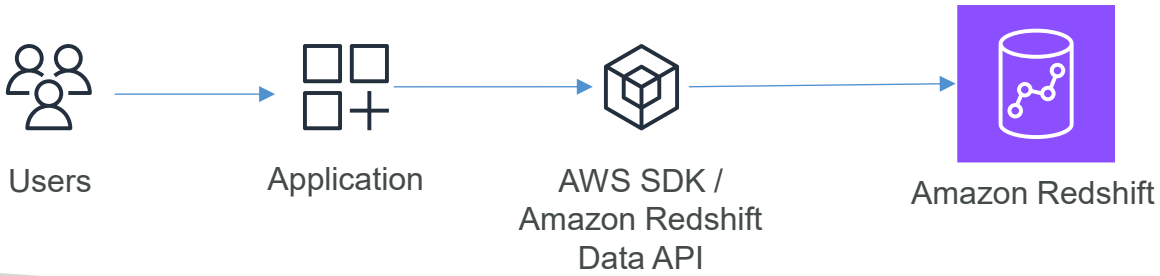
Redshift Data API

- Secure HTTP endpoint for SQL statements to Redshift clusters
  - Provisioned or serverless
  - Individual or batch queries
- Asynchronous
- Does not require managing connections
  - No drivers needed
- Passwords not sent via API
  - Uses AWS Secrets Manager or temporary credentials
- Can be invoked by AWS SDK
  - C++, Go, Java, JavaScript, .NET, Node.js, PHP, Python, Ruby
- AWS CloudTrail captures API calls



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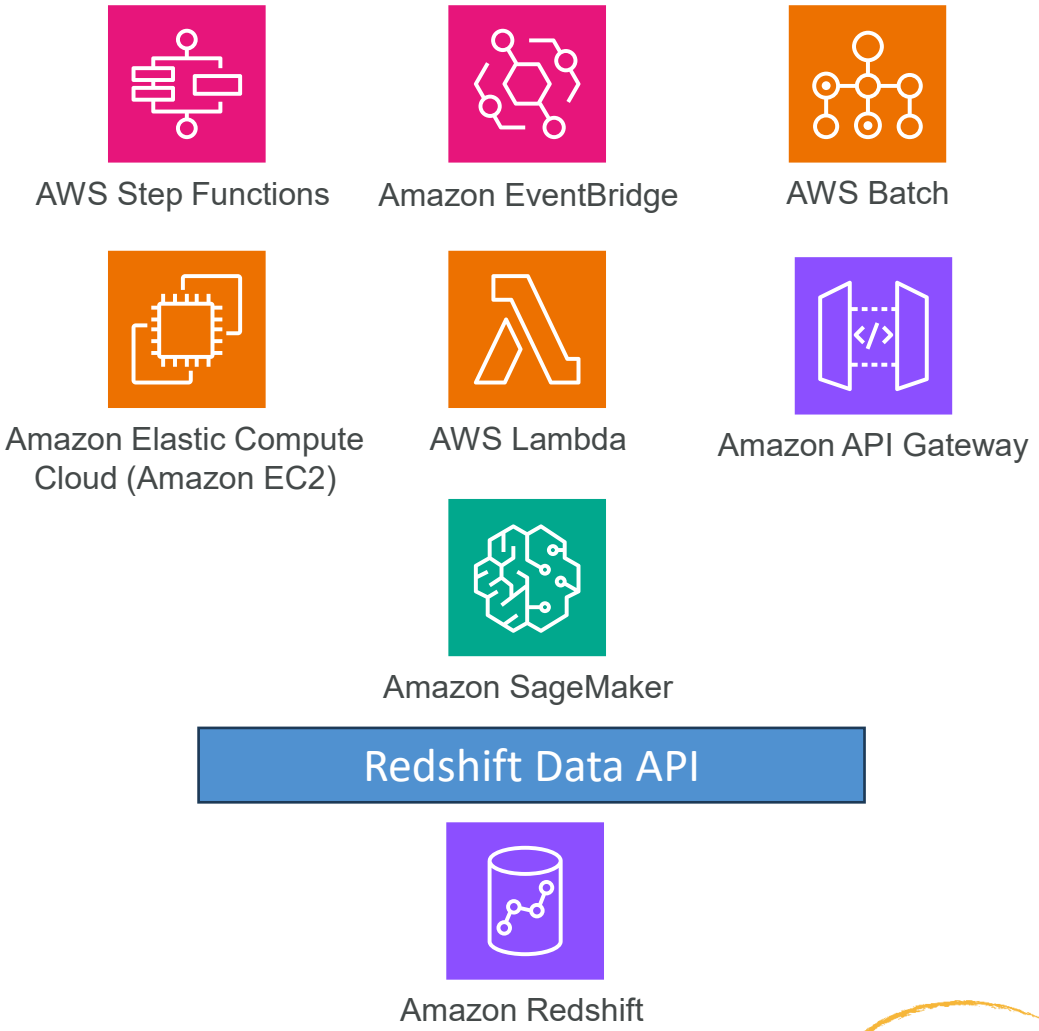
Redshift Data API: Other use cases

- Application integration
  - REST endpoints
- It can also integrate with a variety of other services
  - ETL Orchestration with AWS Step Functions
    - Serverless data processing workflows
    - Event-driven ETL
  - Access from SageMaker notebooks
- Use with Amazon EventBridge
  - Stream data from application to Lambda
    - Set WithEvent parameter to true
  - Schedule Data API operations



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Redshift Data API: Finer Points

- Maximum…
  - Query duration: 24 hours
  - Active queries: 500
  - Query result size (gzip'ed): 100 MB
  - Result retention time: 24 hours
  - Query statement size: 100 KB
  - Packet for query (data per row) : 64 KB
  - Client token retention time: 8 hours
  - Transaction per Second quotas per API (30 for ExecuteStatement)
- API Calls
  - ExecuteStatement, BatchExecuteStatement
  - DescribeStatement, DescribeTable
  - GetStatementResult, CancelStatement
  - Users must have same IAM role or permissions to operate on a given statement
- Cluster must be in a VPC

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